

 FIR/Testing-Outdoor/516800/2023/5963		FIR VEHICLE TESTING		Date: 22/05/2023		
Part No / Group: 51685256R		Specification no :- " 259051680015 "			Action by	Failure class

Subject: -	Performance evaluation of Signa 3525.T Cx RDE in comparison with Ashok Leyland 3520.T BSVI and Signa 3518.T BSVI base vehicle.		
➤ <u>Conclusion:</u> ➤ <u>Signa 3525.T Cx RDE in comparison with Ashok Leyland 3520.T BSVI</u> • Overall performance of Signa 3525.T Cx BSVI RDE is better as compared to Ashok Leyland 3520.T BSVI. PALS position achieved “C” against “C” ➤ <u>Signa 3525.T Cx RDE in comparison with Signa 3518.T BSVI</u> • Overall performance of Signa 3525.T Cx BSVI RDE is better as compared to Signa 3518.T BSVI base vehicle.	--	--	
1.1 <u>Acceleration:</u>			
➤ <u>Signa 3525.T Cx RDE in comparison with Ashok Leyland 3520.T BSVI</u> • Acceleration (0-60 kmph) of Signa 3525.T Cx BSVI RDE was found to be 56.28 Sec. Signa 3525.T BSVI RDE is 9.20% better as compared to Ashok Leyland 3520.T BSVI vehicle and meets PALS target. • Acceleration (0-1000m) of Signa 3525.T Cx BSVI RDE was found to be 75.04 Sec. Signa 3525.T Cx BSVI RDE is 7.44% better as compared to Ashok Leyland 3520.T BSVI vehicle and meets PALS target.	--	--	
➤ <u>Signa 3525.T Cx RDE in comparison with Signa 3518.T BSVI</u> • Acceleration (0-60 kmph) of Signa 3525.T Cx BSVI RDE is 11.40% better as compared to Signa 3518.T BSVI vehicle. • Acceleration (0-1000m) of Signa 3525.T Cx BSVI RDE is 9.52% better as compared to Signa 3518.T BSVI vehicle.	--	--	
1.2 <u>Wide open throttle drivability:</u>	--	--	
➤ <u>Signa 3525.T Cx RDE in comparison with Ashok Leyland 3518.T BSVI</u> • In gear drivability of Signa 3525.T Cx BSVI RDE vehicle was found to be 4.81 kmph/s, 2.34 kmph/s, 2.18 kmph/s, 1.31 kmph/s, 0.91 kmph/s, and 0.64 kmph/s in 1st , 2nd , 3rd , 4th , 5th and 6th gear respectively.			
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 FIR/Testing-Outdoor/516800/2023/5963		FIR VEHICLE TESTING	Date: 22/05/2023	
Part No / Group: 51685256R	Specification no :- " 259051680015 "		Action by	Failure class
➤ Rate of acceleration of Signa 3525.T Cx BSVI RDE vehicle is superior to Ashok Leyland 3520.T BSVI in all gears. ➤ <u>Signa 3525.T Cx RDE in comparison with Signa 3518.T BSVI</u> - Rate of acceleration of Signa 3525.T Cx BSVI RDE vehicle is superior to Signa 3518.T BSVI in all gears. 1.3 <u>Bathtub drivability:</u> ➤ <u>Signa 3525.T RDE in comparison with Ashok Leyland 3520.T</u> - Bathtub drivability of Signa 3525.T Cx BSVI RDE vehicle was found better as compared to Ashok Leyland 3520.T BSVI vehicle in all gears and meets PALS target. ➤ <u>Signa 3525.T BSVI RDE in comparison with Signa 3518.T BSVI</u> - Bathtub drivability of Signa 3525.T Cx BSVI RDE vehicle was found superior to Signa 3518.T BSVI vehicle in all gears. 1.4 <u>Sustain and Restart Gradability:</u> ➤ <u>Signa 3525.T Cx BSVI RDE in comparison with Ashok Leyland 3520.T BSVI</u> - Sustain gradability of Signa 3525.T Cx BSVI RDE was found to be 27.84 %. Signa 3525.T Cx BSVI RDE is 23.40% superior as compared to Ashok Leyland 3520.T BSVI vehicle. - Restart gradability of Signa 3525.T Cx BSVI RDE was found to be 20.3%. Signa 3525.T Cx BSVI RDE is 9.14% superior as compared to Ashok Leyland 3520.T BSVI vehicle. ➤ <u>Signa 3525.T Cx BSVI RDE in comparison with Signa 3518.T BSVI</u> ➤ Sustain gradability of Signa 3525.T Cx BSVI RDE is 14.10% superior as compared to Signa 3518.T BSVI base vehicle. ➤ Restart gradability of Signa 3525.T Cx BSVI RDE is 2.53% superior as compared to Signa 3518.T BSVI base vehicle.			--	--
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 FIR/Testing-Outdoor/516800/2023/5963		FIR VEHICLE TESTING		Date: 22/05/2023		
Part No / Group: 51685256R		Specification no :- " 259051680015 "			Action by	Failure class

2.0 Result: Wide observation.

FAILURES CLASS:

- A: Failure that happen without warning. Detrimental to vehicle / personnel, road users and safety. Can cause death / hospitalization. Noncompliance of regulatory requirement.
- B: Vehicle gets crippled with sufficient warning. Minor injury to occupant & road users (First aid only). Premature failures / repeated failures. Leads to consequential catastrophic damages. Need immediate attention.
- C: Customer irritants. Capable of damaging aesthetics. Failure can be attended at next servicing Schedule.
- D: Customer perception issues and minor irritants.

3.0 Test Vehicle:

1. Signa 3525.T Cx BSVI RDE (ERC – 1203)
Chassis No. – PC73553022RJ00737
Engine No. – 12M64182675

Input Doc Ref. : Testing-Outdoor/15047

DID No : NA

DDP No : NA

Enclosures:

- Annexure 1: Panel chart
Annexure 2: PAT Sheet
Annexure 3: PAT Summary
Annexure 4: Test request
Annexure 5: Test plan
Annexure 6: EOL

Work Order No. :- P.21.EA.458

Make :- Tata Motors Ltd

T I No. – 41965

Earlier Ref: -

- FIR_Testing Outdoor_507900_2020_2052_dated 17_03_2020_Performance evaluation on Ashok Leyland 4825 BSVI vehicle.

4.0 Project Status: DR3

5.0 Introduction:

- As a part of RDE program, Signa 3525.T Cx BSVI RDE vehicle was offered for performance evaluation, Ashok Leyland 3520.T BSVI was taken as core benchmark and Signa 3518.T BSVI vehicle was taken as Base for performance comparison. This report will reveal detail of this trial with the status.

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FIR

VEHICLE TESTING

Date: 22/05/2023

Part No / Group: 51685256R

Specification no :- " 259051680015 "

Action
byFailure
class**6.0 Testing**

➤ Following test was conducted during trials.

Sr. No.	Test Description	Standard Referred	Instrument used
1	Acceleration	TST_VT_WG_0002	VBOX SX with GPS, Weighbridge (CRS Gate)
2	WOT Drivability	TST_VT_WG_0002	
3	Bath -Tub Drivability	TST_VT_WG_0261	
4	Gradability	TST_VT_WG_11	
5	Restart Gradability	TST_VT_WG_127	
6	Coastdown	TST_VT_WG_36	

7.0 Vehicle details:

Sr. No.	Parameters	Base Vehicles BSVI	Benchmark Vehicle BSVI	Test Vehicles BSVI
		Signa 3518.T Ashwamedh BSVI	Ashok Leyland 3520	Signa 3525.T RDE BSVI
1	Engine	6 Cylinder 5.6 L / 180 HP BSVI	6 Cylinder 5.7L / 200 HP/ BSVI	6 Cylinder 6.7 L / 250 HP BSVI
1.1	Engine Max. Power	180 hp @ 2300 RPM	200 hp @ 2400 RPM	250 HP @ 2300 rpm
1.2	Engine Max. Torque	850 Nm @ 1000-1600 RPM	700 Nm @ 1200-2000 RPM	950 Nm @ 1000-1800 rpm
1.3	Power to weight (HP/Ton)	5.14	5.71	6.76
1.4	Torque to weight (Nm/Ton)	24.29	20	25.68
1.5	Engine Calibration Number	XHTBA63082		
1.6	Engine Calibration Maturity	90%		90%
1.7	Emission Cal Maturity	100%		
1.8	Regeneration Strategy	Auto	Yes	Auto
1.9	Urea Dosing	Yes	Yes	Yes
2.0	Exhaust Downpipe Insulation	Updated downpipe (2.1)		Updated downpipe (2.1)
2.1	EOL Parameters	LBSC- 1700,1700 rpm @35000 kg ; VAM- 2.5,2.5 kmph/s @ 10, 40 kmph ; GDP- 45, 45 kmph	NA	LBSC- 1700,1700 rpm @35000 kg ; VAM- 2.5,2.5 kmph/s @ 10, 40 kmph ; GDP- 45, 45 kmph, RSG – 80 Kmph
2	Fuel Economy Improvement Technologies			
2.1	Engine Cooling Fan	E-Viscous	E-Viscous	E-Viscous
2.2	Fuel Economy Switch	Yes (3 Mode)		Yes (3 Mode)
2.3	Gear Shift Adviser	Yes		Yes
2.4	Accelerator Pedal Map	Yes		Yes
3	Other Parameters			
3.1	Fuel Tank type	Polymer	Polymer	Polymer
3.2	Driveline	G 950 DD, FGR 9.36	Eaton (6 speed), FGR 9.01	G 950 DD, FGR 9.36
3.3	Gear Ratio	I-9.36, II-5.2, III-3.05, IV-1.84, V-1.33, VI-1,	I-9.01, II-5.27, III-3.22, IV-2.04, V-1.36, VI-1	I-9.36, II-5.2, III-3.05, IV-1.84, V-1.33, VI-1,
3.4	RAR	5.12	5.83	5.12
3.5	Tire size	Front – Rib 295/90R20 Apollo Rear – Lug 295/90R20 Apollo	Front – Rib 295/90R20 Apollo Rear – Lug 295/90R20 Apollo	Front – Rib 295/90R20 Bridgestone Rear – Lug 295/90R20 Bridgestone

Vehicle Photographs



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FIR/Testing-Outdoor/516800/2023/5963

FIR

VEHICLE TESTING

Date: 22/05/2023

Part No / Group: 51685256R

Specification no :- " 259051680015 "

Action
byFailure
class**8.0 Torque table:-****➤ Balance mode (TF – 100%)**

	600	800	1000	1200	1400	1600	1800	2000	2200	2400	2600	2800
1	800	800	950	950	735	655	655	572	510	503	119	0
1.33	800	800	950	950	735	655	655	572	510	503	119	0
1.84	800	800	950	950	735	655	655	572	510	503	119	0
3.05	800	800	950	950	735	655	655	516	510	503	119	0
5.2	800	800	950	950	950	950	950	890	809	580	119	0
9.36	800	800	950	950	950	950	950	890	809	580	119	0

➤ Economy mode (TF – 100%)

	600	800	1000	1200	1400	1600	1800	2000	2200	2400	2600	2800
1	800	800	950	650	495	495	495	450	417	396	119	0
1.33	800	800	950	650	554	554	554	513	463	440	119	0
1.84	800	800	950	650	554	554	554	513	463	440	119	0
3.05	800	800	950	684	606	606	606	513	463	440	119	0
5.2	800	800	950	950	950	950	950	890	809	580	119	0
9.36	800	800	950	950	950	950	950	890	809	580	119	0

➤ 9.0 Observation:-

Overall performance Signa 3525.T Cx BSVI RDE is better as compared to Ashok Leyland 3520.T BSVI and PALS position achieved "C" against "C".

10. Future testing:- As per further development.

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Annexure - 1

Level 1 Attribute: PAT- Performance and Drivability					PANEL chart Specification: Max Power : 250 hp @2300 rpm Max Torque : 950 Nm @ 1000 - 1800 rpm Driveline : G950 DD, 5.12 RAR , Bridgestone 295/90R20 EOL: RSG-80, VAM 1.5 Kmph/sec, 1.5 Kmph/sec, 10 kmph, 40 Kmph, LBSC 1700, 1700 RPM & GDP- 45 Kmph, 45 Kmph					VE CVBU/ PD/00 Panel Chart No.: VE CVBU/ M&HCV/ Signa 3525.T Cx BSVI RDE Atulya/ PD/00																						
Programme / MY : Signa 4830.T					Gateway : DR3		Owner: Pavan Sindgikar		Project Name: Signa 3525.T Cx BSVI RDE					Date: 17/05/2023																		
Level 1 Summary:												Statement Of Intent (SOI) or DNA Description:																				
<table><tr><td rowspan="3">PALS</td><td rowspan="3">Level 1:</td><td>PALS</td><td colspan="2">Status</td></tr><tr><td>Current/ Gateway</td><td>Job#1 Predicted</td></tr><tr><td>C</td><td>C</td></tr></table>												PALS	Level 1:	PALS	Status		Current/ Gateway	Job#1 Predicted	C	C	Performance: Haulage application Fuel Economy: Haulage application											
PALS	Level 1:	PALS	Status																													
		Current/ Gateway	Job#1 Predicted																													
		C	C																													

4 Deriv.	Attribute Target Summary:							Current Target	Status	Base Vehicle	Competitor Data BS VI		
	Level 2	Level 3	Level 4	Level 5	Level 6	PALS	Metric	w.r.t Ashok Leyland 3520.T BSVI	Current/Gateway Signa 3525.T Cx RDE BSVI	Signa 3518.T BSVI	Core Benchmark Ashok Leyland 3520.T	Adjecent Benchmark	Remarks
									Heavy mode	Heavy mode	--	--	
	Specification								LBSC - 1700,1700,35000 Kg VAM- 1.5 Kmph/s, 1.5 Kmph/s, 10 kmph, 40kmph GDP- 45 kmph, 45 kmph RSG - 80kmph	LBSC - 1700,1700,35000 Kg VAM- 2.5 Kmph/s, 2.5 Kmph/s, 10 kmph, 40kmph GDP- 50 kmph, 50 kmph RSG - 80kmph	--	--	--
	Vehicle GVW (Payload)						kg		35000	35000	35000	--	
	Power (Max) @ rpm						HP		250 hp @ 2300 rpm	180 hp @ 2300 rpm	200 hp @ 2400 RPM	--	
	Power to Weight ratio						HP / Ton		7.14	5.14	5.71	--	
	Torque (Max) @ rpm						Nm		950 Nm @ 1000-1800 rpm	850 Nm @ 1000-1600 rpm	700 Nm at 1200-2000 rpm	--	
	Torque to Weight ratio						Nm / Ton		27.14	24.29	20.00	--	
	Tyres size						~		Bridgestone 295/90 R20	Apollo 295/90 R20	Apollo JK Tyres 295/90 R20	--	
Gear Box Ratios						~		G950 DD, FGR -9.36	G950 DD, FGR -9.36	Eaton 6 speed , FGR - 9.01	--		
Final Drive Ratio						~		5.12	5.12	5.83	--		
Acceleration at rated load					C				C				
0-60 kmph						Sec		≤ 61.98	57.99	63.52	61.98	--	
0-1000 m						Sec		≤ 81.08	76.15	82.94	81.08	--	
Bathtub Drivability (Time) at Rated load					UC				C				
3rd Gear													
10 - 15 Kmph						sec		≤ 2.96	2.55	4.55	2.96	--	
15 - 20 Kmph						sec		≤ 2.55	2.53	3.26	2.55	--	
4th Gear													
20 - 25 Kmph						sec		≤ 4.13	3.94	6.43	4.13	--	
25 - 30 Kmph						sec		≤ 3.63	3.59	4.55	3.63	--	
30 - 35 Kmph						sec		≤ 4.2	3.71	4.23	4.20	--	
35 - 40 kmph						sec			4.99				
5th Gear													
25 - 30 Kmph						sec		≤ 6.64	5.76	8.50	6.64	--	
30 - 35 Kmph						sec		≤ 6.45	5.39	6.90	6.45	--	
35 - 40 Kmph						sec		≤ 6.4	5.45	6.11	6.40	--	
6th Gear													
30-40 Kmph						sec		≤ 20.97	16.37	27.37	20.97	--	
40-50 Kmph						sec		≤ 20.15	15.00	18.00	20.15	--	
50-60 Kmph						sec		≤ 21.41	15.52	17.46	21.41	--	
Gradeability at rated load					AL				C				
Sustain Gradeability						%		≥ 22.56	27.84	24.40	22.56	--	
Restart Gradeability						%		≥ 18.6	20.30	19.80	18.60	--	

Executive Summary:		1. Ashok Leyland 3520.T BSVI have been taken for target setting. 2. Signa 3525.T BSVI RDE was having 95% engine calibration maturation. 3.Target has been taken at-par as per Cx variant approach																									
Tested by:														Verified and Approved by:													
Debabrata Das/Palash Das/ Swapnil Jadhav PAT Engineer														Ajaykumar Jadhav / Sanjeev Kumar PAT Lead / Project Lead													



Annexure 2

Document Revision No.: VE CVBU/ M&HCV/ Signa 3525.T Cx RDE
Atulya/ 00

Project Name: Signa 3525.T Atulya Cx BSVI RDE

Aggregate and test parameter							Unit	Test Type	Classification (IDCI)	Ref.Standard / Work Guideline	Compliance	Base Vehicle	Core Benchmark	Target	Test Vehicle	Status (PTC)	Remarks	CoC Remark
Sr.No.	Level 2	Level 3	Level 4	Level 5	Level 6	Level 7						Signa 3518.T 5.6L Aswamedh BSVI G950 DD 6S RAR 5.12	Ashok Leyland 3520.T Eaton 6 Speed DD RAR 5.83	Signa 3525.T Cx RDE BSVI G950 DD RAR 5.12				
A Performance Evaluation																		
Heading												LBSC - 1700,1700,35000 Kg VAM- 2.5 Km/hrs, 2.5 Km/hrs, 10 km/h, 40 km/h GDP- 40 km/h, 45 km/h R50 - 80km/h	--	--	LBSC - 1700,1700,35000 Kg VAM- 1.5 Km/hrs, 1.5 Km/hrs, 10 km/h, 40 km/h GDP- 40 km/h, 45 km/h R50 - 80km/h	--	1. Ashok Leyland 3520.T BSVI have been taken for target setting. 2. Signa 3525.T BSVI RDE was having 55% engine calibration sign off report.	
S1	Minimum speed						Heading	Heavy Mode				Heavy Mode	--					
S1.1	1st gear						Kmph	Specification				3.1	2.3		3.0	--		
S1.2	2nd gear						Kmph	Specification				5.5	3.8		5.2	--		
S1.3	3rd gear						Kmph	Specification				9.4	6.5		9.2	--		
S1.4	4th gear						Kmph	Specification				15.6	10.6		15.3	--		
S1.5	5th gear						Kmph	Specification				21.6	16.2		22.6	--		
S1.6	6th gear						Kmph	Specification				28.7	22.3		28.4	--		
S1.7	7th gear						Kmph	Specification				NA	NA		NA	--		
S1.8	8th gear						Kmph	Specification				NA	NA		NA	--		
S2	Maximum speed						Heading							--				
S2.1	1st gear						Kmph	Specification				7.8	10.1		7.5	--		
S2.2	2nd gear						Kmph	Specification				14.0	17.2		13.3	--		
S2.3	3rd gear						Kmph	Specification				23.8	27.5		23.3	--		
S2.4	4th gear						Kmph	Specification				39.5	43.5		43.0	--		
S2.5	5th gear						Kmph	Specification				50.3	64.2		45.0	--		
S2.6	6th gear						Kmph	Specification				80.0	80.0		80.0	--		
S2.7	7th gear						Kmph	Specification				NA	NA		NA	--		
S2.8	8th gear						Kmph	Specification				NA	NA		NA	--		
S2.9	Vehicle maximum speed						Kmph	Specification				80.0	80	80	80	OK		
S3	Acceleration						Heading							--				
S3.1	AC Off						Sec	Heading							OK			
S3.2.1	0-60 km/h (0.37 mph)						Sec	Target				63.52	61.08	61.08	57.89	OK		
S3.2.4	0-1000 m						Sec	Target				82.94	61.08	61.08	76.15	OK		
S3	Cruis ability						Heading							--				
S3.2	AC Off						%	Heading							--			
S3.2.1	Maximum (sustained) Gradability (Forward)						%	Target				24.4	22.56	22.56	27.84	OK		
S3.2.2	Max Restart Gradability (Forward)						%	Target				19.8	18.6	218.6	20.39	OK		
S4	Coast down (km/h²)						Heading							--				
S4.2	Laden condition						Kwh	Heading						--				
S4.2.1	Power consumed at 40 km/h						W	Data Generation				35.6	36.6	--	26.6	NA		
S4.2.2	a						W	Data Generation				1889.76	2239	--	1840	NA		Signa 3525.T is better than base and core benchmark vehicle
S4.2.3	b						W	Data Generation				0.34	0.29	--	0.31	NA		
S4.2.4	Coeff of rolling resistance						mm	Data Generation				0.0055	0.0061	--	0.0061	NA		
B Drivability Evaluation																		
Heading															--			
S5	WOT Drivability (AC ON)						Heading							--				
S5.1	1st gear 4-6 Km/h						Sec	Data Generation				1.11	1.09	--	0.62	--		
S5.2	2nd gear 8-12 Km/h						Sec	Data Generation				2.17	2.22	--	1.71	--		
S5.3	3rd gear 12-20 Km/h						Sec	Data Generation				5.38	4.44	--	3.67	--		
S5.4	4th gear 19-33 Km/h						Sec	Data Generation				12.66	11.49	--	10.72	--		
S5.5	5th gear 25-42 Km/h						Sec	Data Generation				22.46	23.05	--	18.78	--		
S5.6	6th gear 36-50 km/h						Sec	Data Generation				29.71	29.34	--	21.87	--		
S5.10	AC Off (Rate of acceleration)						Sec	Heading						--				
S5.11	1st gear 4-6 Km/h						Kmph/sec	Data Generation				1.80	1.83	--	3.23	--		
S5.12	2nd gear 8-12 Km/h						Kmph/sec	Data Generation				1.84	1.80	--	2.34	--		
S5.13	3rd gear 12-20 Km/h						Kmph/sec	Data Generation				1.49	1.80	--	2.18	--		
S5.14	4th gear 19-33 Km/h						Kmph/sec	Data Generation				1.11	1.27	--	1.31	--		
S5.15	5th gear 25-42 Km/h						Kmph/sec	Data Generation				0.76	0.74	--	0.91	--		
S5.16	6th gear 36-50 km/h						Kmph/sec	Data Generation				0.54	0.48	--	0.64	--		
S6	Bathub Curve - Part Throttle Drivability (AC ON)						Heading							--				
S6.7	2nd gear						Heading											
S6.13	3rd gear						Heading											
S6.14	10-15 km/h						sec	Target				4.56	2.96	4.236	2.95	OK		
S6.15	15-20 km/h						sec	Target				3.26	2.55	4.255	2.51	OK		
S6.1	4th gear						Heading											
S6.4	20-25 km/h						sec	Target				6.43	4.13	5 4.13	3.94	OK		
S6.5	25-30 km/h						sec	Target				4.55	3.63	5 3.63	3.69	OK		
S6.6	30-35 km/h						sec	Target				4.23	4.20	5 4.2	3.71	OK		
S6.7	35-40 km/h						sec	Target							4.99	NA		
S6.8	5th gear						Heading											
S6.9	25-30 km/h						sec	Target				8.50	6.64	5 6.64	5.76	OK		
S6.10	30-35 km/h						sec	Target				6.90	6.45	5 6.45	5.59	OK		
S6.11	35-40 km/h						sec	Target				6.11	6.40	5 6.4	5.45	OK		
S6.13	6th gear						Heading											
S6.14	30-40 km/h						sec	Target				27.37	20.97	5 20.97	16.37	OK		
S6.9	40-50 km/h						sec	Target				18.00	20.15	5 20.15	15.00	OK		
S6.15	50-60 km/h						sec	Target				17.46	21.43	5 21.41	15.52	OK		
C Engine Calibration Activities																		
Heading															--			
1.0 System Verification Trials																		
S1	Powertrain Characteristics (AC ON)						Heading							--	--	--	NA	
S1.1	Powertrain Characteristics (AC OFF)						Target							--	Pass	Pass	OK	
2.0 Vehicle Performance Trials																		
S1	Vehicle Speed detection						Heading							--	Pass	Pass	OK	
S2	Engine Speed detection						Target							--	Pass	Pass	OK	
S2.1	Gear detection						Target							--	Pass	Pass	OK	
S2.4	Driver idle						Target							--	Pass	Pass	OK	
3.0 Engine derate verification																		
S2	Limp home validation						Heading							--	Pass	Pass	OK	
S2.1	Reduced torque fuel validation						Heading							--	Pass	Pass	OK	
S2.3	Evaluation of Take-off ability, Acceleration and Deceleration						Target							--	Pass	Pass	OK	
D Vehicle feature validation and Fuel Economy features / End of Line (EOL) parameters segregation and verification																		
Heading															--			
S9	Vehicle Mass Estimation						Target						--	Pass		OK		
S9	Road Slope Estimation						Target						--	Pass		OK		
S9	Vehicle acceleration management (VAM tuning)						Target						--	Pass		OK		
S9	Electronic vehicle speed Limiter (EVSL) / Load based speed control (LBSC)						Target						--	Pass		OK		
S9	Vehicle speed limiter (Top and one-gear down)						Target						--	Pass		OK		
S9	EVSL+ VAM / LBSC+VAM						Target						--	Pass		OK		
S9	Gear Down Protection (GDP)						Target						--	Pass		OK		
S9	Idle shutdown						Target						--	Pass		OK		
S9	Average fuel economy (AFE)						Target						--	Pass		OK		
S9	IFE						Target						--	Pass		OK		
S9	Engine Brake control						Target						--	Pass		OK		
S9	Cruise control (CC)						Target						--	Pass		OK		
S9	Power Take off (PTO)						Target						--	Pass		OK		
S9	Remote PTO						Target						--	Pass		OK		
S9	Road Speed governor						Target						--	Pass		OK		
S9	Engine Protection shutdown						Target						--	Pass		OK		
S9	Powertrain protection						Target						--	Pass		OK		
S9	Trip-Info						Target						--	Pass		OK		

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A20 F411 (R1) May, 2021

Sr.No.	Aggregate and test parameter						Unit	Test	Classification @DR3	Ref.Standard / Work Guideline	Simulation	Base Vehicle		Core Benchmark	Target	Test Vehicle		Status (RYG)	Remarks	CoC Remark				
	Level 2	Level 3	Level 4	Level 5	Level 6	Level 7						Signa 3518.T 5.6L Aswamedh BSVI G850 DD 65 RAR 5.12	Ashok Leyland 3520.T Eaton 6 Speed DD RAR 5.83	Signa 3525.T Cx RDE BSVI G850 DD RAR 5.12										
E	Environmental Trials Verification																			Heading		--		
1.0	Summer trials																			Heading		--		
1.4	Hot start									Target					--	Pass	Pass	Pass	Pass					
1.5	Low idle governor calibration									Target					--	Pass	Pass	Pass	Pass	Carry over from OBD I validation				
1.6	Driven idle									Target					--	Pass	Pass	Pass	Pass					
1.7	Subjective Drivability Assessment									Target					--	Pass	Pass	Pass	Pass					
2.0	Winter trials																			Heading		--		
2.2	Cold Start (at Ambient as per Application requirement)									Target					--	Pass	Pass	Pass	Pass					
2.3	Smoke level (visually) during free acceleration after start									Target					--	Pass	Pass	Pass	Pass					
2.4	Subjective Drivability Assessment									Target					--	Pass	Pass	Pass	Pass					
3.0	High altitude trials/ calibration																			Heading		--		
3.2	Low idle governor calibration						rpm		New Development	Target						--	Pass	Pass	Pass	Pass				
3.3.1	% Deterioration in Drivability with base calibration						%			Target						--	Pass	Pass	Pass	Pass				
3.10.1	Turbocharger duty cycle mapping									Target						--	Pass	Pass	Pass	Pass				
3.10.1.1	Smoke during driving present or not									Target						--	Pass	Pass	Pass	Pass				
3.10.1.2	Assessment of turbocharger Whistling noise								New Development	Target						--	Pass	Pass	Pass	Pass				
3.11.1	Power calculation						%			Target						--	Pass	Pass	Pass	Pass				
F	Onboard Diagnostics																			Heading		--		
5.0	OBD Diagnostic - TML tool verification									Heading					--	Pass	Pass	Pass	Pass	Carry over from OBD I validation				
5.9	Deviate Strategies for Critical Faults									Heading					--	Pass	Pass	Pass	Pass					
6.0	Verification of OBD interface for Comprehensive Component Monitors									Heading					--	Pass	Pass	Pass	Pass					
Sign Off															Index for Status									
Responsibility							Name							Signature					Off target, no resolution					
Prepared by: PAT Engineer							Debabrata Das/Palash Das / Swapnil												Marginally off target, resolution planned/low risk					
Checked & Approved by: PAT Lead / Project Lead							Ajaykumar Jadhav / Sanjeev kumar												Met target, no concern					

PED evaluation of Signa 3525.T RDE BSVI in comparison with Ashok Leyland 3520 BSVI and Signa 3518.T Ashwamedh BSVI

PED| 17.04.23

OUR CULTURE CREDO

AT TATA MOTORS

We are connecting aspirations by being bold in thought and action, owning every opportunity and challenge, Solving together as one team and engaging all our stakeholders with empathy.
We are **MORE WHEN ONE!**

BE BOLD
Taking calculated **risk** is key to making progress. We act with confidence and **agility** to accomplish our goals

OWN IT
Feeling and acting **empowered** is critical to drive results. We have an **Owner's Mind-set** and each of us takes full responsibility for the outcomes

SOLVE TOGETHER
Leveraging our collective genius while holding each other **accountable** helps us deliver the best. We **collaborate** proactively and transparently to achieve innovative solutions

BE EMPATHETIC
Embracing diversity makes us stronger for differences are opportunities to learn. We work with **passion to delight customers** and deliver greater success to our stakeholders

- ✓ Vehicle specification comparison
- ✓ Performance Data Summary
- ✓ Minimum and Maximum Gear Speed
- ✓ Wide open throttle performance
- ✓ Tractive effort comparison
- ✓ Acceleration
- ✓ WOT drivability
- ✓ Bathtub drivability
- ✓ Coast Down Data
- ✓ Sustain and restart gradeability

Vehicle Specification Comparison

Annexure - 3

Sr. No.	Parameters	Base Vehicles BSVI	Benchmark Vehicle BSVI	Test Vehicles BSVI
		Signa 3518.T Ashwamedh BSVI	Ashok Leyland 3520	Signa 3525.T RDE BSVI
1	Engine	6 Cylinder 5.6 L / 180 HP BSVI	6 Cylinder 5.7L / 200 HP/ BSVI	6 Cylinder 6.7 L / 250 HP BSVI
1.1	Engine Max. Power	180 hp @ 2300 RPM	200 hp @ 2400 RPM	250 HP @ 2300 rpm
1.2	Engine Max. Torque	850 Nm @ 1000-1600 RPM	700 Nm @ 1200-2000 RPM	950 Nm @ 1000-1800 rpm
1.3	Power to weight (HP/Ton)	5.14	5.71	6.76
1.4	Torque to weight (Nm/Ton)	24.29	20	25.68
1.5	Engine Calibration Number	XHTBA63082		
1.6	Engine Calibration Maturity	90%		90%
1.7	Emission Cal Maturity	100%		
1.8	Regeneration Strategy	Auto	Yes	Auto
1.9	Urea Dosing	Yes	Yes	Yes
2.0	Exhaust Downpipe Insulation	Updated downpipe (2.1)		Updated downpipe (2.1)
2.1	EOL Parameters	LBSC- 1700,1700 rpm @35000 kg ; VAM- 2.5,2.5 kmph/s @ 10, 40 kmph ; GDP- 45, 45 kmph	NA	LBSC- 1700,1700 rpm @35000 kg ; VAM- 2.5,2.5 kmph/s @ 10, 40 kmph ; GDP- 45, 45 kmph, RSG – 80 Kmph
2	Fuel Economy Improvement Technologies			
2.1	Engine Cooling Fan	E-Viscous	E-Viscous	E-Viscous
2.2	Fuel Economy Switch	Yes (3 Mode)		Yes (3 Mode)
2.3	Gear Shift Adviser	Yes		Yes
2.4	Accelerator Pedal Map	Yes		Yes
3	Other Parameters			
3.1	Fuel Tank type	Polymer	Polymer	Polymer
3.2	Driveline	G 950 DD , FGR 9.36	Eaton (6 speed), FGR 9.01	G 950 DD , FGR 9.36
3.3	Gear Ratio	I-9.36, II-5.2, III-3.05, IV-1.84, V-1.33, VI-1,	I-9.01, II-5.27, III-3.22, IV-2.04, V-1.36, VI-1	I-9.36, II-5.2, III-3.05, IV-1.84, V-1.33, VI-1,
3.4	RAR	5.12	5.83	5.12
3.5	Enclosed with FIR/Testing-Outdoor/ Tire size	516800/2023/5963 Front – Rib 295/90R 20 Apollo Rear – Lug 295/90R20 Apollo	Front – Rib 295/90R 20 Apollo Rear – Lug 295/90R20 Apollo	Front – Rib 295/90R 20 Bridgestone Rear – Lug 295/90R20 Bridgestone

Performance Data Summary

Annexure - 3

Parameter	Unit	Target	PALS	Benchmark	Base Vehicle	Test Vehicle (Extrapolated)			RDE Status (%) w.r.t Benchmark	RDE Status (%) w.r.t Base	PALS status
				Ashok Leyland 3520 BSVI Eaton	Signa 3518.T BSVI HM	Signa 3525.T BSVI RDE HM	Signa 3525.T BSVI RDE BM	Signa 3525.T BSVI RDE LM			
Acceleration at Rated Load											
0 - 60 kmph	sec	≤ 61.98	C	61.98	63.52	56.28	77.95	67.18 (0 - 50 Kmph)	9.20%	11.40%	
0 - 1000 m	sec	≤ 82.94	C	81.08	82.94	75.04	81.78	89.88	7.44%	9.52%	
Driveability(Rate of acceleration) at Rated Load											
1st Gear(4-6)	sec	Not Applicable/ Data Generation	Not Applicable/ Data Generation	1.09	1.11	0.62	0.80	0.85	Not Applicable/ Data Generation	Not Applicable/ Data Generation	--
2nd Gear(8-12)	sec			2.22	2.17	1.71	1.72	1.75			
3rd Gear(12-20)	sec			4.44	5.38	3.67	3.90	3.70			
4th Gear(19-33)	sec			11.49	12.66	10.72	14.49	17.15			
5th Gear(25-42)	sec			25.54	38.29	18.78	25.89	36.38			
6th Gear(36-50)	sec			50.57	32.38	21.87	30.76	48.04			
1st Gear(4-6)	kmph/sec	Not Applicable/ Data Generation	Not Applicable/ Data Generation	1.84	1.90	4.81	3.73	3.52	Not Applicable/ Data Generation	Not Applicable/ Data Generation	--
2nd Gear(8-12)	kmph/sec			1.81	1.84	2.34	2.33	2.29			
3rd Gear(12-20)	kmph/sec			1.80	1.49	2.18	2.05	2.16			
4th Gear(19-33)	kmph/sec			1.22	1.11	1.31	0.97	0.82			
5th Gear(25-42)	kmph/sec			0.74	0.76	0.91	0.66	0.47			
6th Gear(36-50)	kmph/sec			0.48	0.54	0.64	0.46	0.29			
Bath-Tub Driveability											
3rd Gear											
10 --15	Sec	≤ 2.96	UC	2.96	4.55	2.55	2.83	2.88	13.93%	44.01%	
15 - 20	Sec	≤ 2.55	UC	2.55	3.26	2.53	2.71	2.26	0.96%	22.53%	
4th Gear											
20 - 25	Sec	≤ 4.13	UC	4.13	6.43	3.96	5.26	5.96	4.16%	38.44%	
25 - 30	Sec	≤ 3.63	UC	3.63	4.55	3.61	5.12	6.39	0.64%	20.73%	
30 - 35	Sec	≤ 4.20	UC	4.20	4.23	3.73	4.88	6.37	11.18%	11.81%	
35 - 40	Sec	--				5.01	4.57	4.34			
5th Gear											
25 - 30	Sec	≤ 6.64	UC	6.64	8.5	5.79	6.59	9.10	12.82%	31.90%	
30 - 35	Sec	≤ 6.45	UC	6.45	6.90	5.41	7.63	11.37	16.14%	21.61%	
35 - 40	Sec	≤ 4.13	UC	6.40	6.11	5.47	7.38	11.87	14.48%	10.42%	
6th Gear											
30 - 40	Sec	≤ 20.97	UC	20.97	27.37	17.18	19.48	24.51	18.07%	37.23%	
40 - 50	Sec	≤ 20.15	UC	20.15	18.00	15.60	22.80	35.62	22.61%	13.36%	
50 - 60	Sec	≤ 21.41	UC	21.41	17.46	16.23	25.18	42.55	24.18%	7.05%	
60 - 70	Sec	--				17.89	28.98				
Sustained and Restart Gradeability											
Sustained Gradeability	%	≥22.56	AL	22.56	24.4	27.84	--	--	23.40%	14.10%	
Restart Gradeability	%	≥18.6	AL	18.6	19.8	20.3	--	--	9.14%	2.53%	

Signal 3525.T Cx BSVI RDE Performance. PALS position achieved "AL" against "C"

Acceleration

- Acceleration of Signal 3525.T Cx BSVI RDE was found to be **better as compared to target** in 0-60 kmph speed and 0-1000 meter band

Bath tub Drivability

- Drivability of Signal 3525.T Cx BSVI RDE was found to be **better as compared to target** in 3rd, 4th, 5th and 6th gear.

Sustain & Restart Gradeability

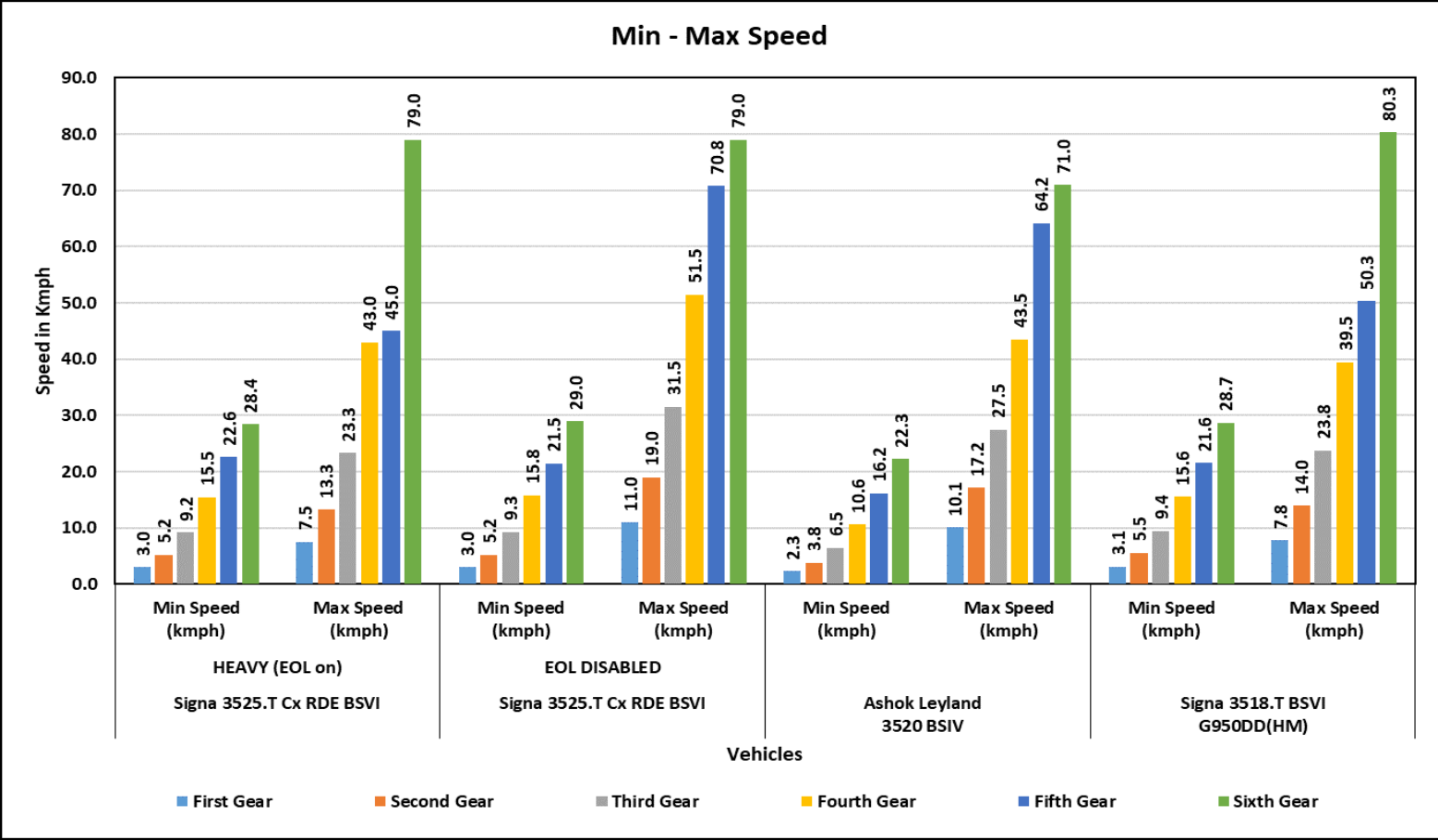
- Restart Gradability of 3525.T Cx RDE BSVI was found better than target.
- Sustain Gradability of 3525.T Cx RDE BSVI was found better that target

Note: As the all tests was done with IAL. Signal 3525.T Cx RDE BSVI Data was extrapolated from Signal 3725.T IAL

Enclosed with FIR/Testing-Outdoor/516800/2023/5953

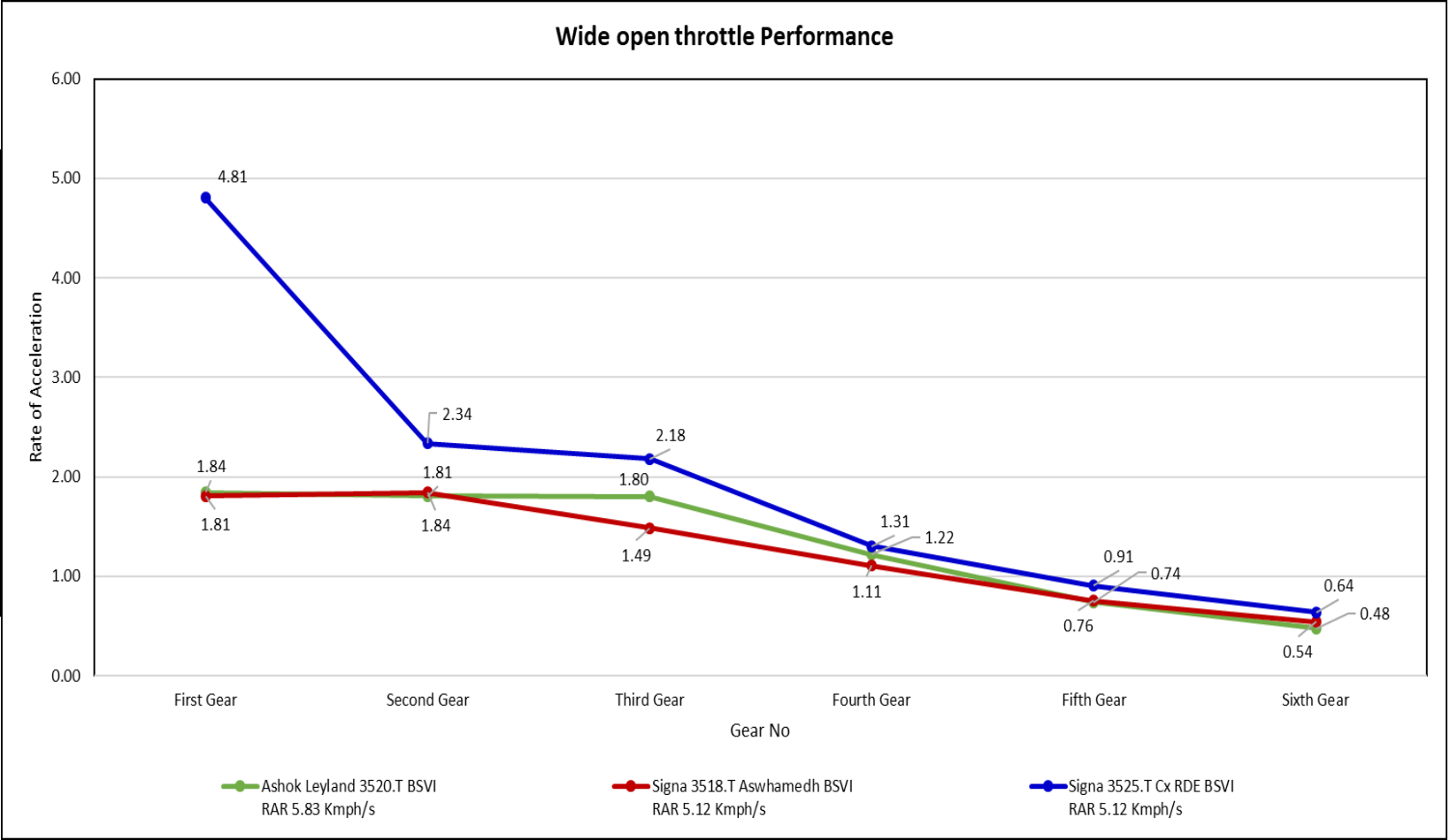
Minimum & Maximum Gear Wise speed Comparison

Test Vehicle					Benchmark Vehicle		Base Vehicle	
GEAR	Signa 3525.T Cx RDE BSVI		Signa 3525.T Cx RDE BSVI		Ashok Leyland 3520 BSIV		Signa 3518.T BSVI G950DD(HM)	
	HEAVY (EOL on)		EOL DISABLED					
	Min Speed (kmph)	Max Speed (kmph)	Min Speed (kmph)	Max Speed (kmph)	Min Speed (kmph)	Max Speed (kmph)	Min Speed (kmph)	Max Speed (kmph)
1	3.0	7.5	3.0	11.0	2.3	10.1	3.1	7.8
2	5.2	13.3	5.2	19.0	3.8	17.2	5.5	14.0
3	9.2	23.3	9.3	31.5	6.5	27.5	9.4	23.8
4	15.5	43.0	15.8	51.5	10.6	43.5	15.6	39.5
5	22.6	45.0	21.5	70.8	16.2	64.2	21.6	50.3
6	28.4	79.0	29.0	79.0	22.3	71.0	28.7	80.3

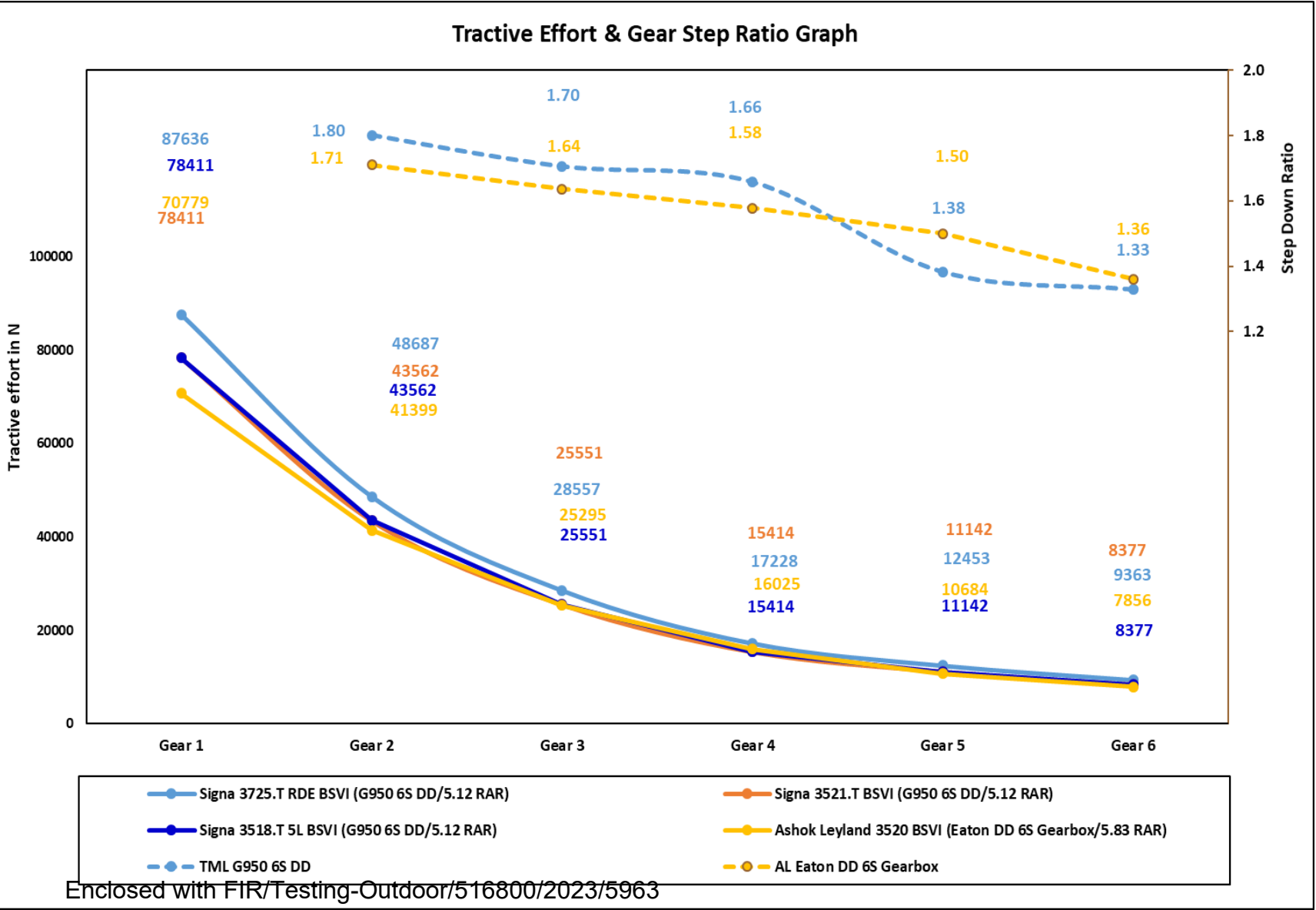


Wide open throttle Performance

Gear No	Ashok Leyland 3520.T BSVI RAR 5.83	Signa 3518.T Aswhamedh BSVI RAR 5.12	Signa 3525.T Cx RDE BSVI RAR 5.12
	Kmph/s	Kmph/s	Kmph/s
First Gear	1.84	1.81	4.81
Second Gear	1.81	1.84	2.34
Third Gear	1.80	1.49	2.18
Fourth Gear	1.22	1.11	1.31
Fifth Gear	0.74	0.76	0.91
Sixth Gear	0.48	0.54	0.64



Tractive effort comparison with Benchmark

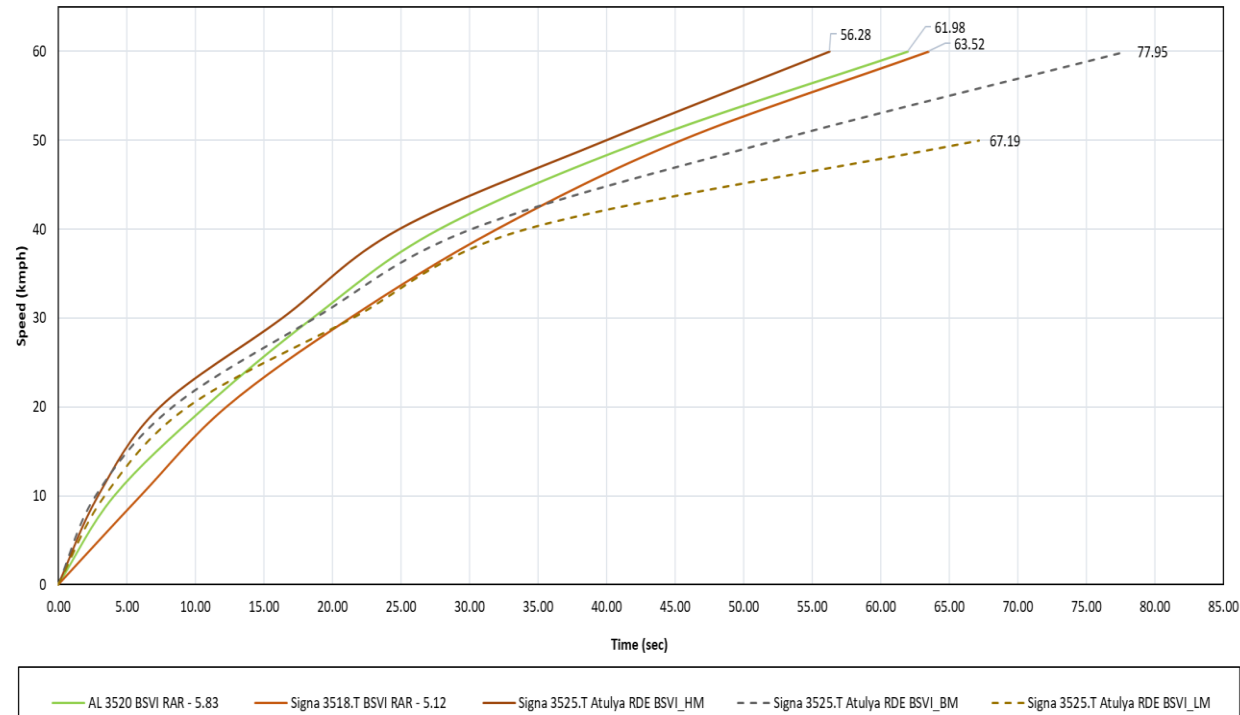


Remarks:

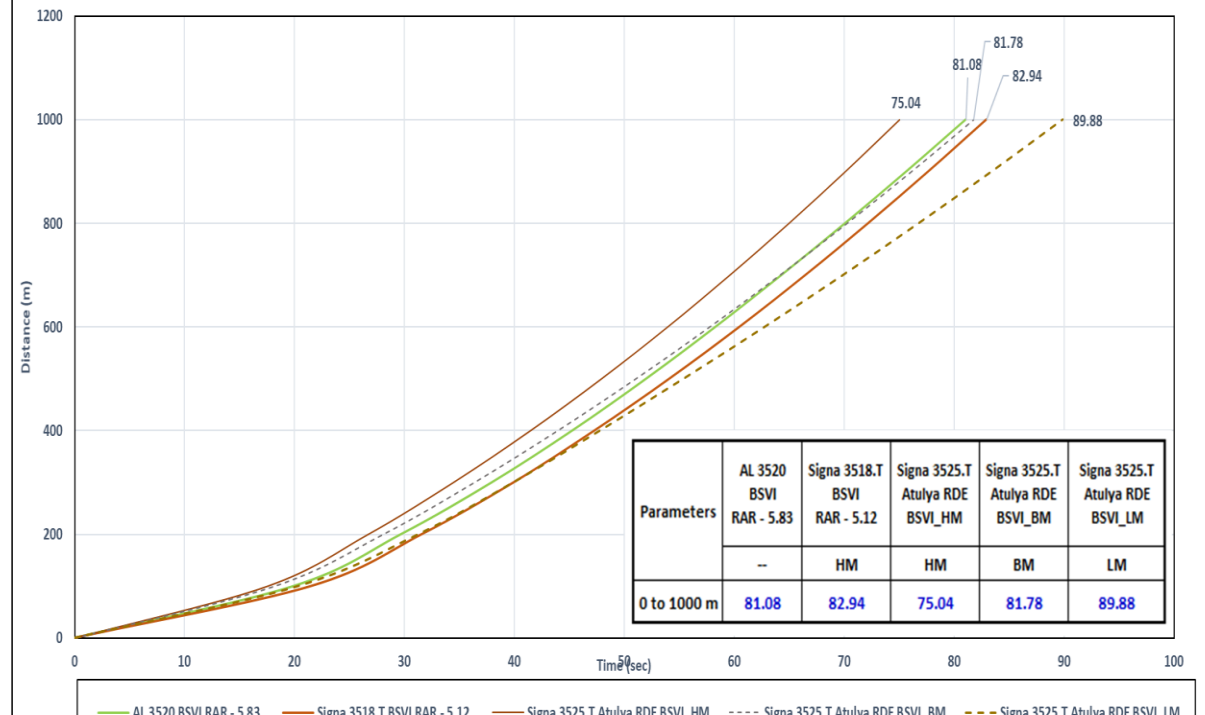
- Tractive effort of Signa 3525.T RDE BSVI is Superior than Benchmark & Base Vehicle vehicle in all gear.

Acceleration Performance – Through Gear

Acceleration 0 to 60kmph

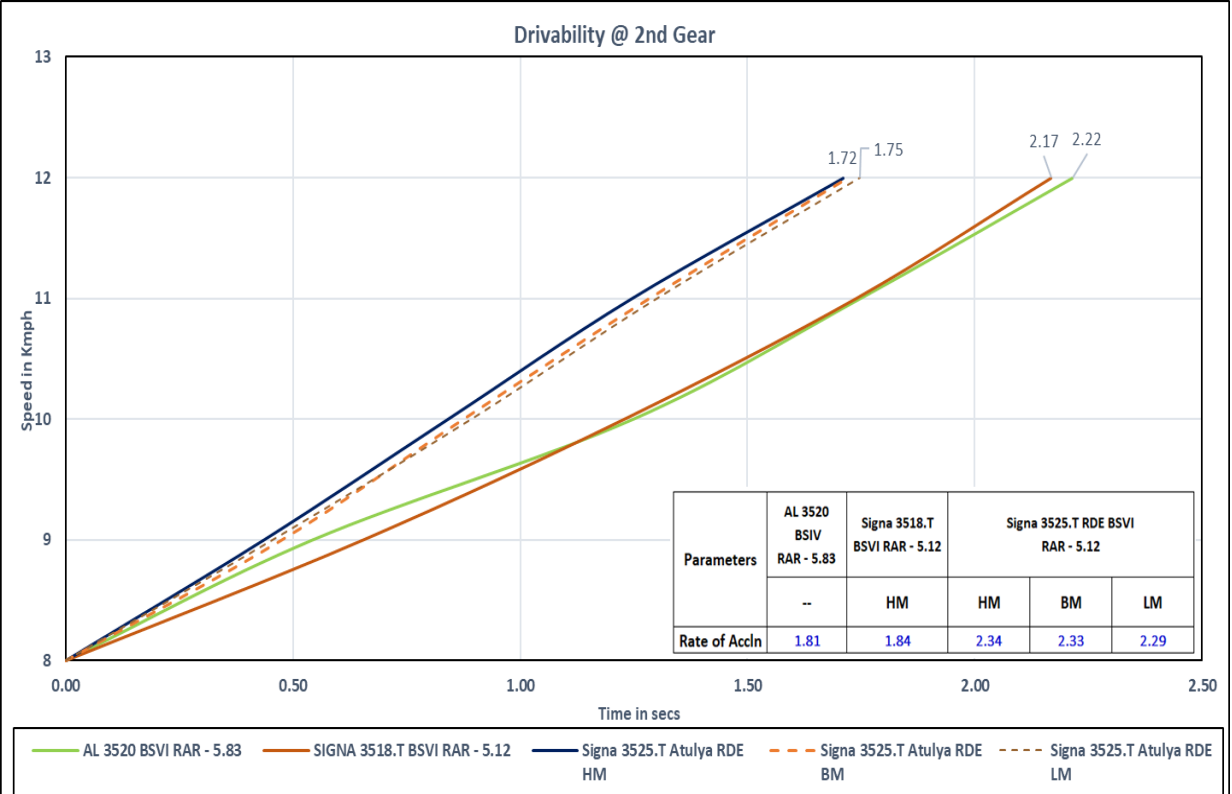
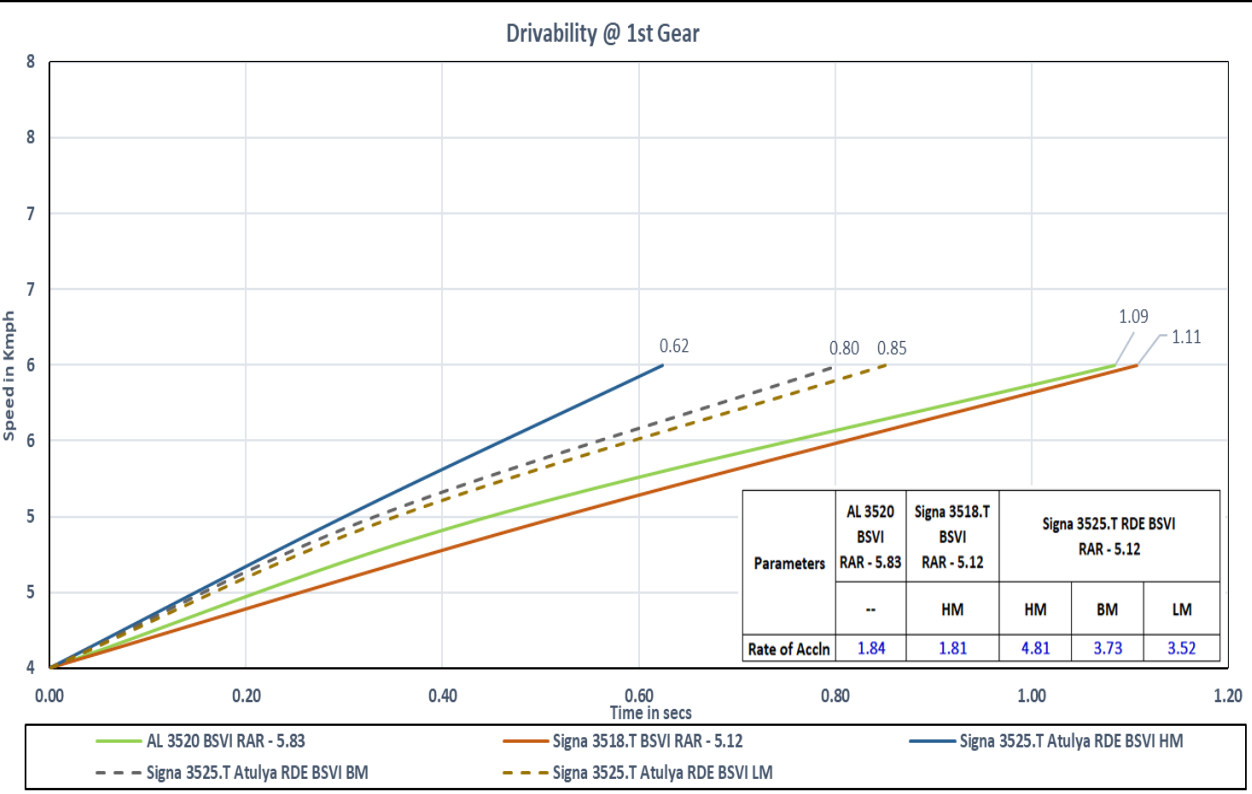


Acceleration 0 to 1000m



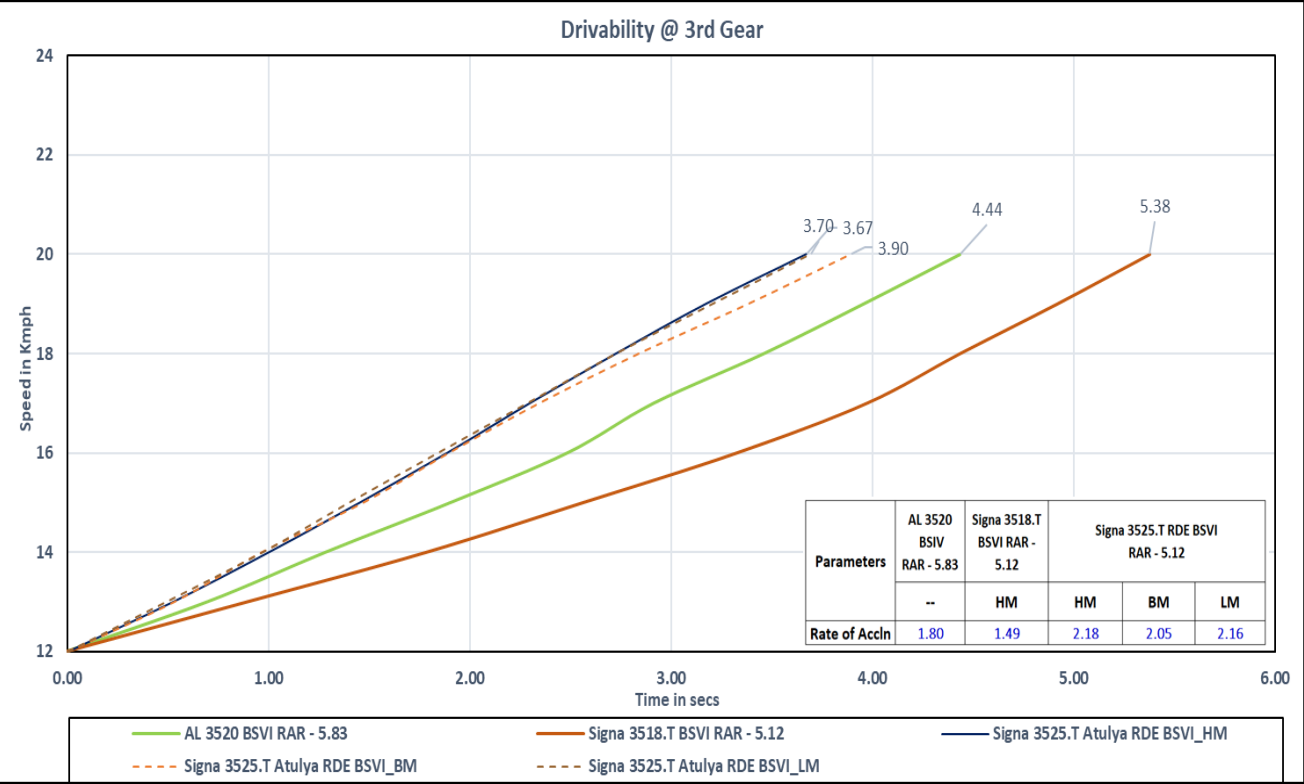
- ❖ Acceleration performance through gear for Signa 3525.T Cx BSVI RDE **is better than** AL 3520.T and Signa 3518.T BSVI in 0 – 60 Kmph
- ❖ Acceleration performance through gear for Signa 3525.T Cx BSVI RDE **is better than** AL 3520.T and Signa 3518.T BSVI in 0 – 1000 meters

WOT Drivability @ 1st & 2nd Gear



- ❖ In First gear, WOT acceleration performance of Signa 3525.T Cx RDE vehicle is better as compared with AL 3520.T BSVI and Signa 3518.T BSVI
 - ❖ In Second gear, WOT acceleration performance of Signa 3525.T Cx RDE vehicle is better as compared with AL 3520.T BSVI and Signa 3518.T BSVI
- Image Credit: F&V Testing - Outdoor/516800/2023/5963

WOT Drivability @ 3rd & 4th Gear



- ❖ In third gear, WOT acceleration performance of Signa 3525.T Cx RDE vehicle **is better as compared** with AL 3520.T BSVI and Signa 3518.T BSVI
 - ❖ In fourth gear, WOT acceleration performance of Signa 3525.T Cx RDE vehicle **is better as compared** with AL 3520.T BSVI and Signa 3518.T BSVI
- Enclosed With NAM Testing-Outdoor/516800/2023/5963

WOT Drivability @ 5th & 6th Gear



- ❖ In fifth gear, WOT acceleration performance of Signa 3525.T Cx RDE vehicle **is better as compared** with AL 3520.T BSVI and Signa 3518.T BSVI
 - ❖ In sixth gear, WOT acceleration performance of Signa 3525.T Cx RDE vehicle **is better as compared** with AL 3520.T BSVI and Signa 3518.T BSVI
- Enclosed with NAM Testing-Outdoor/516800/2023/5963

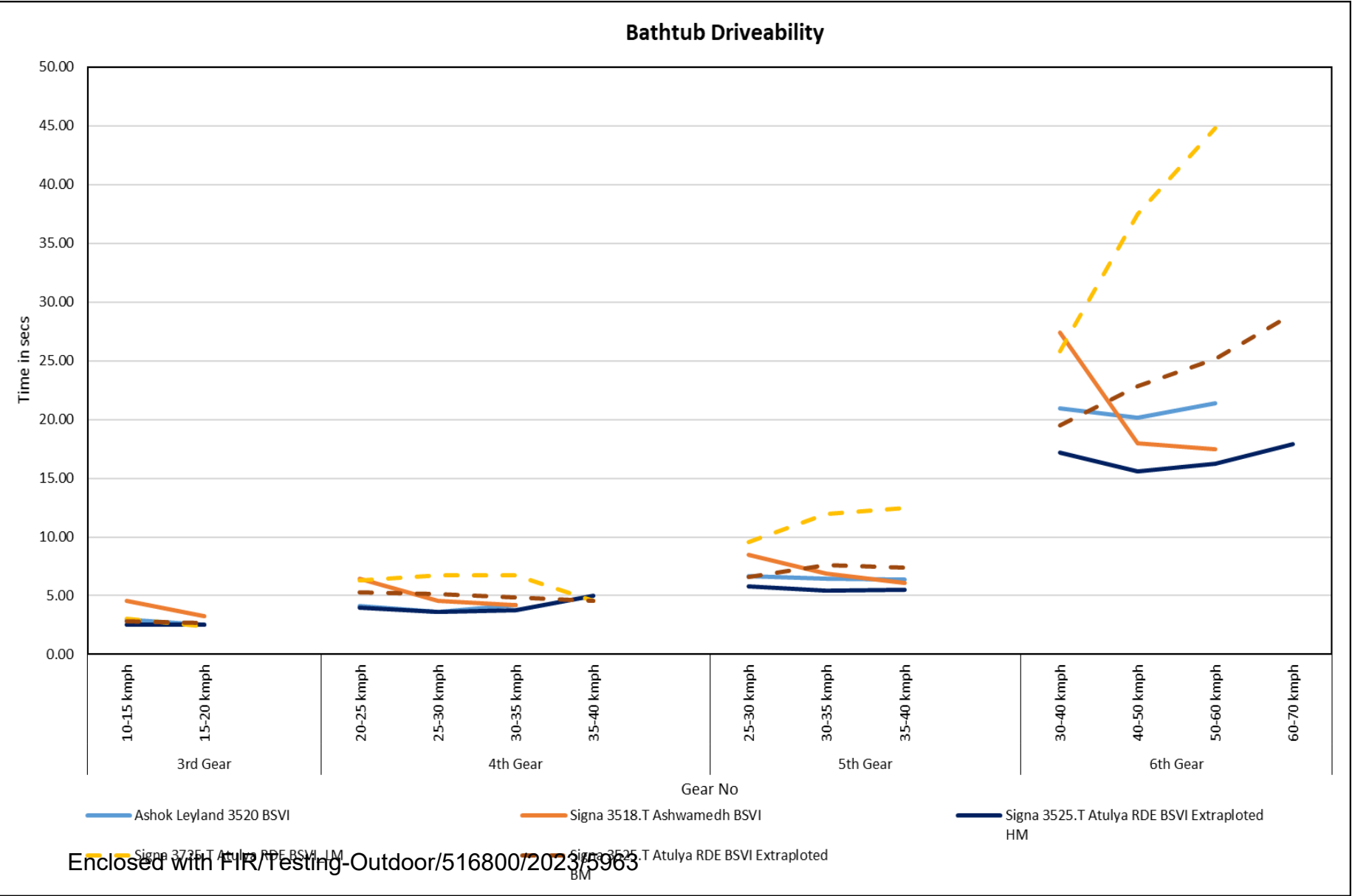
Bathtub Drivability Report

Parameter	Unit	Target	PALS	Benchmark	Base Vehicle	Test Vehicle (Extrapolated)			RDE Status (%) w.r.t Benchmark	RDE Status (%) w.r.t Base	PALS status
				Ashok Leyland 3520 BSVI Eaton	Signa 3518.T BSVI HM	Signa 3525.T BSVI RDE HM	Signa 3525.T BSVI RDE BM	Signa 3525.T BSVI RDE LM			
Bath-Tub Driveability											
3rd Gear											
10 --15	Sec	≤ 2.96	UC	2.96	4.55	2.55	2.83	2.88	13.93%	44.01%	
15 - 20	Sec	≤ 2.55	UC	2.55	3.26	2.53	2.71	2.26	0.96%	22.53%	
4th Gear											
20 - 25	Sec	≤ 4.13	UC	4.13	6.43	3.96	5.26	5.96	4.16%	38.44%	
25 - 30	Sec	≤ 3.63	UC	3.63	4.55	3.61	5.12	6.39	0.64%	20.73%	
30 - 35	Sec	≤ 4.20	UC	4.20	4.23	3.73	4.88	6.37	11.18%	11.81%	
35 - 40	Sec	--				5.01	4.57	4.34			
5th Gear											
25 - 30	Sec	≤ 6.64	UC	6.64	8.5	5.79	6.59	9.10	12.82%	31.90%	
30 - 35	Sec	≤ 6.45	UC	6.45	6.90	5.41	7.63	11.37	16.14%	21.61%	
35 - 40	Sec	≤ 4.13	UC	6.40	6.11	5.47	7.38	11.87	14.48%	10.42%	
6th Gear											
30 - 40	Sec	≤ 20.97	UC	20.97	27.37	17.18	19.48	24.51	18.07%	37.23%	
40 - 50	Sec	≤ 20.15	UC	20.15	18.00	15.60	22.80	35.62	22.61%	13.36%	
50 - 60	Sec	≤ 21.41	UC	21.41	17.46	16.23	25.18	42.55	24.18%	7.05%	
Enclosed with FR/Testing-Outdoor/516800/2023/5963						17.89	28.98				

Remarks: As the all tests was done with IAL. Signa 3525.T Cx RDE BSVI Data was extrapolated from Signa 3725.T IAL

Vehicle is meeting target in all the gears.

Bathtub Drivability Comparison with Benchmark & Base Vehicles



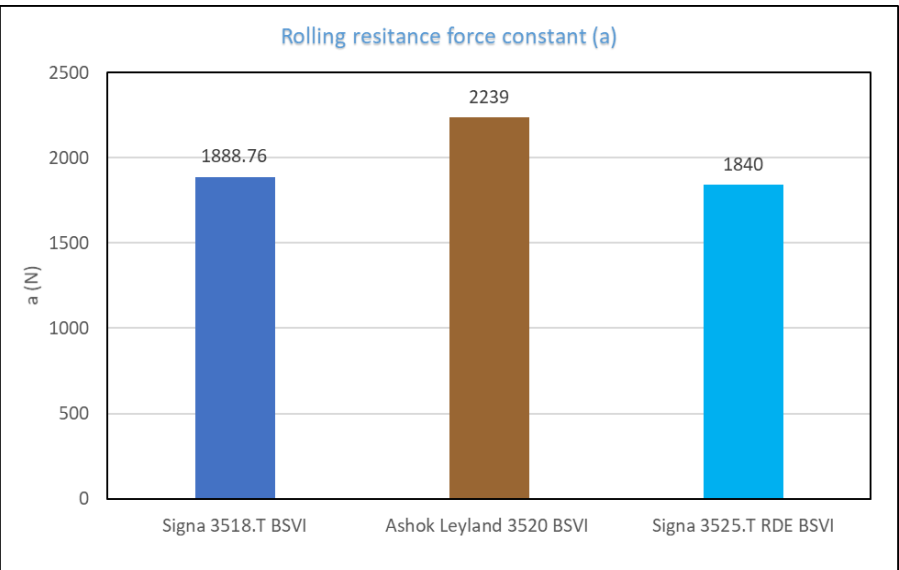
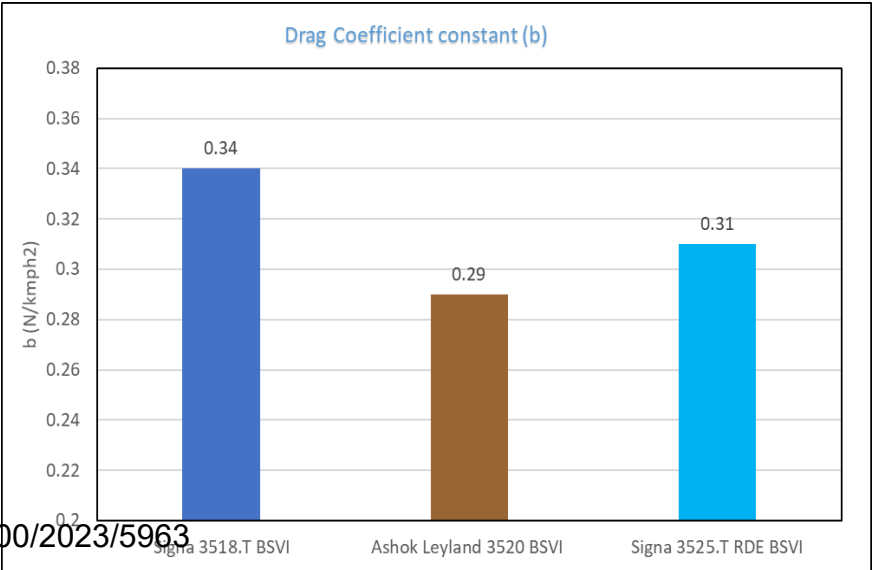
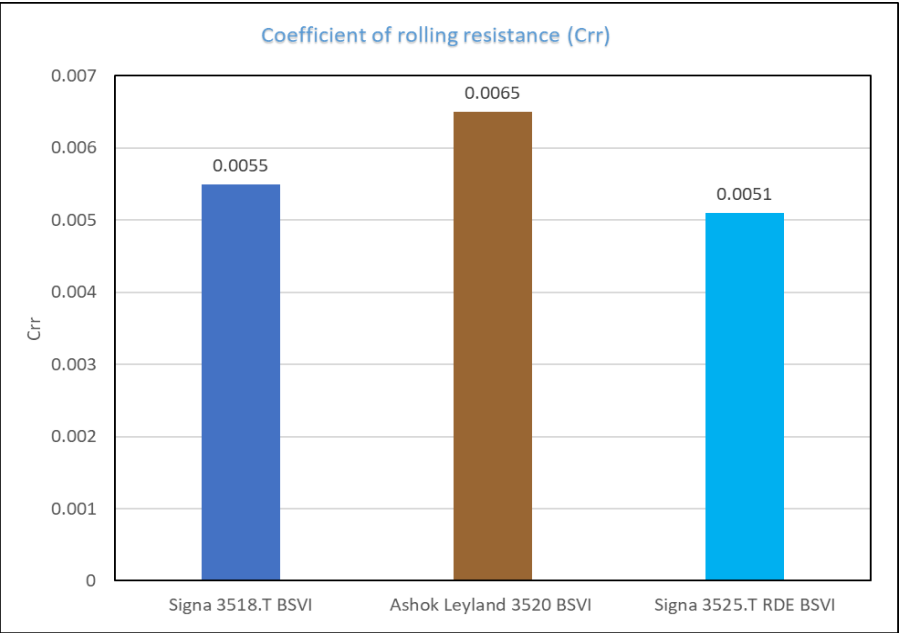
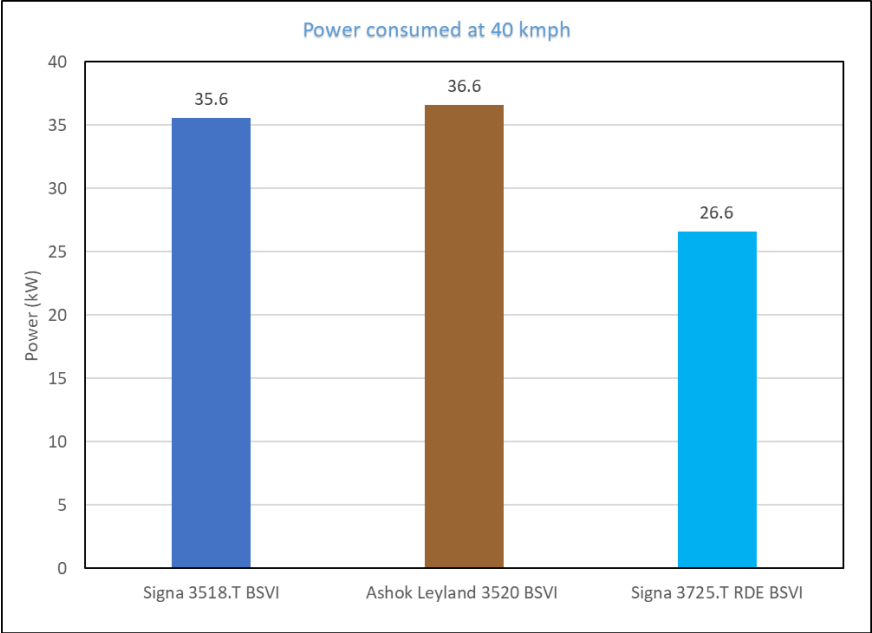
Remarks: As the all tests was done with IAL. Signa 3525.T Cx RDE BSVI Data was extrapolated from Signa 3725.T IAL

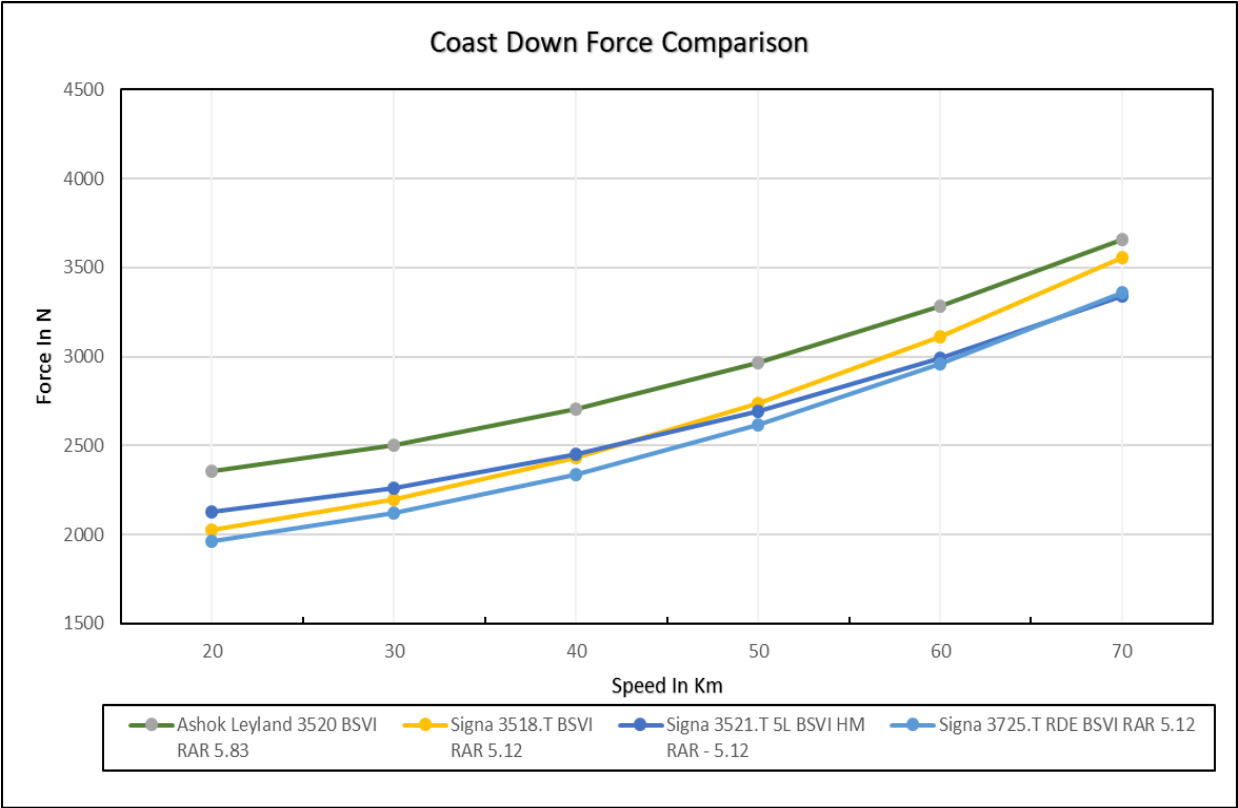
Signa 3525.T vehicle is better than AL 3520 and Signa 3518.T.

Vehicle is meeting PALS target in all gears

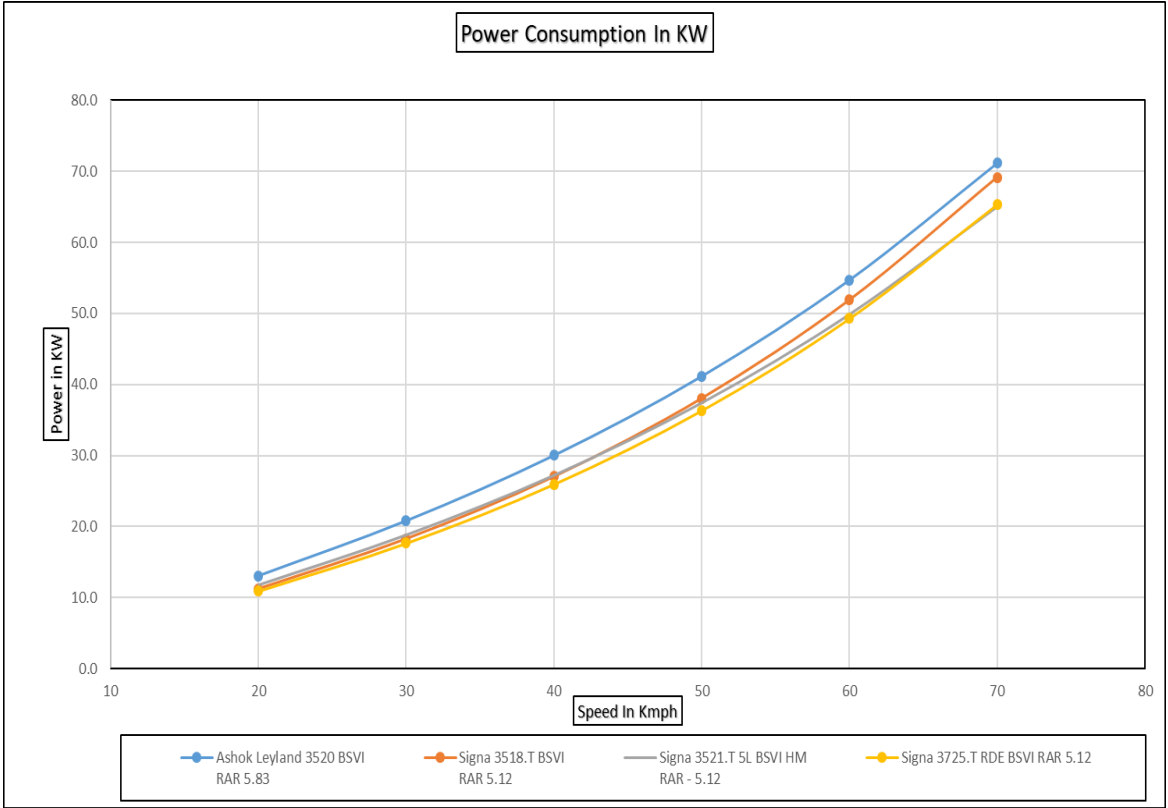
Coast Down Comparison with Benchmark & Base Vehicles

Vehicle	Power consumed at 40 kmph	a	b	Coeff of rolling resistance
	kW	N	N/kmph ²	--
Signal 3518.T BSVI	35.6	1888.76	0.34	0.0055
Ashok Leyland 3520 BSVI	36.6	2239	0.29	0.0065
Signal 3525.T RDE BSVI	26.6	1840	0.31	0.0051



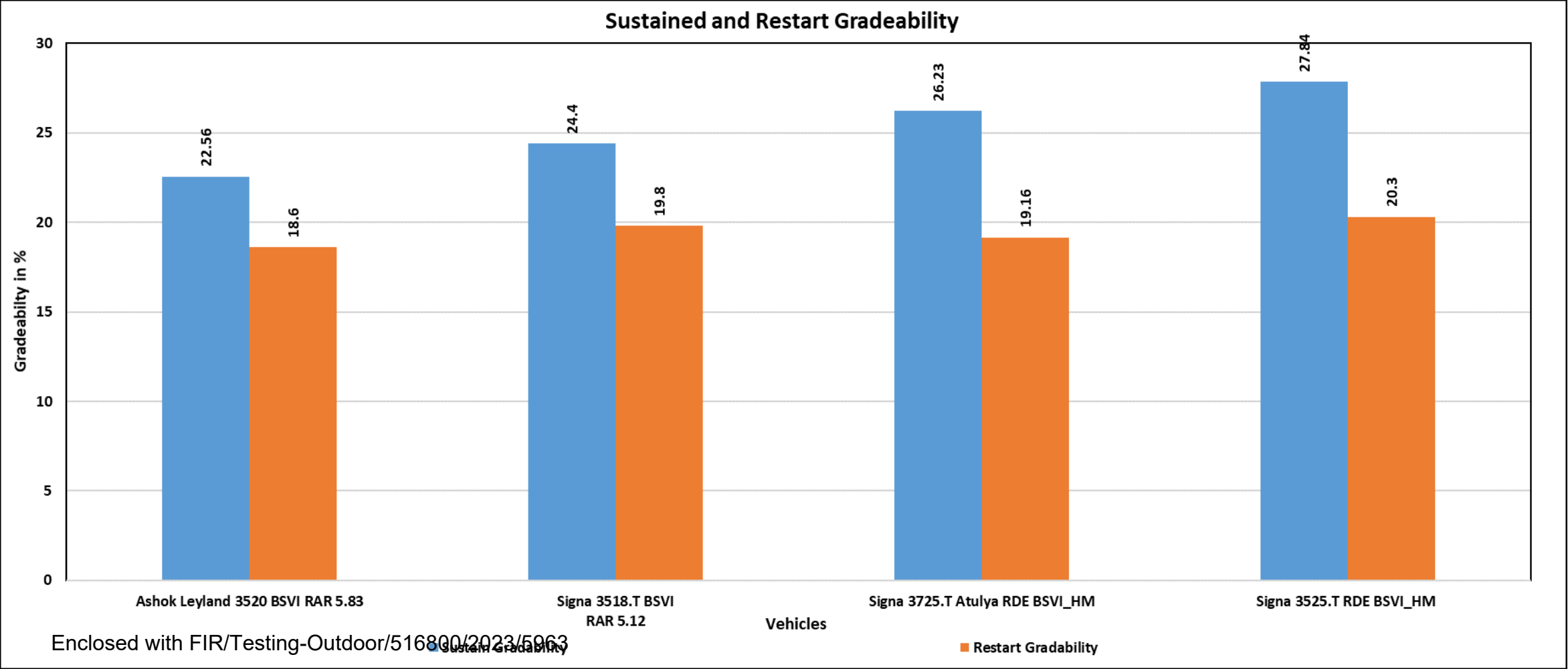


Force Measured for Signa 3725.T RDE is better than Benchmark & Base Vehicle in all Speed



Power Consumption for Signa 3725.T RDE is better than Benchmark & Base Vehicle in all Speed

	Ashok Leyland 3520 BSVI RAR 5.83	Signa 3518.T BSVI RAR 5.12	Signa 3725.T Atulya RDE BSVI_HM	Signa 3525.T RDE BSVI_HM
Sustain Gradability	22.56	24.4	26.23	27.84
Restart Gradability	18.6	19.8	19.16	20.3



Thank You!

TEST REQUISITION

Ref.	NA	Date:	30/03/2023
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To,

ERC Testing

Kindly arrange to take up for testing of the following component.

1	Part Description	TATA SIGNA 3525.T BSVI				
2	Part Number	51685253R,				
3	Drawing number	516800000006,	Mod status	A,2,	Mod Date	11/05/2023
4	Vehicle Model / Project	TATA SIGNA 3525.T RDe BSVI				
5	WBS Number	P.22.EA.501 -RDE Truck 5.6 Ltr 3123.T / 3523.T				
6	Supplier	NA				
7	Number of samples					
8	Inspection report	Enclosed	☒ Yes	☐ No	If No commitment date:	
9	Material report	Enclosed	☐ Yes	☒ No	If No commitment date:	
10	Suppliers test report	Enclosed	☒ Yes	☐ No	If No commitment date:	
11	Reason for testing	First Source Development	☐	Weight Reduction	☐	
		Alternate source development	☐	Cost Reduction	☐	
		Data generation Test	☐	Regulatory test	☐	
		Any other (pl. specify)	☐	New Development	☐	
12	What testing required?	DYNAMIC				
13	Remarks					
14	Peripheral Parts Availability	YES				

Authorized Signature:	
Name:	
Dept:	

Contact Person:	Prithviraj Ray
Phone Number	

For Testing Dept Use only	Test Item Number and Date(stamp)	Test Engineer

TATA MOTORS LIMITED ERC PUNE	TEST PLAN	DATE: 17/05/2023			
	VEHICLE TESTING DEPARTMENT				
Description: TATA SIGNA 3525.T RDe BSVI		Test Item No: 41965			
		New	☒	Moderate	☒
		Major	☒	Minor	☒

Part No: 51685253R,	Modification No: A,2,	Enclosures: Refer FOR
Drawing No: 516800000006,		
Make: NA	WBS Number: P.22.EA.501 -RDE Truck 5.6 Ltr 3123.T / 3523.T	Copies to :
Vehicle Type: TATA SIGNA 3525.T RDe BSVI	No. of Samples: 1	
Test Requested by:	VEHICLE-I	Equipments: Refer FIR
Test Requisition Reference:	Testing-Outdoor/15047	
Reason for Test/s:	Component & System Level Performance Tests	
Regulatory Requirement:	a) Component:	
	b) Vehicle:	

Sr. No.	Test description	Test specification	Acceptance criteria	No. of Samples	Test Responsibility	Proto Phase (Mule/ Alpha/ Beta/PP)	Remarks
1	PD trial	PD trial	As per PALS	1	Debabrata Das	Proto	Completed

REMARKS: PD trial	INTERNAL REF:	
TEST TO FAILURE: No		
MEOST: No		
RELIABILITY TESTING: No		
BENCHMARKING TESTING: No		
DFMEA/FMEA DONE: Yes		
PREPARED BY	CHECKED BY	APPROVED BY
Debabrata Das		

VC No.					VC : 51685156R	VC : 51685256R
Vehicle Model					Signa 3525.T.AC	Signa 3525.T.Blower
Annexure-6					Signa 3525.T.Abiya (G-950 DD [5.12 RAR [295/90R20])	Signa 3525.T.Abiya (G-950 DD [5.12 RAR [295/90R20])
Parameter	VEPS Name	DEFAULT VALUE *	MIN/MAX *	UNITS		
Fuel Economy						
Gear Down Protection (GDP)						
Gear Down Protection Enable	20061	N		Y = Enable N = Disable	Y	Y
Gear Down Protection Heavy Load Vehicle Speed	11440	28	Min: 16 Max: (10000)	mph	45 kmph (27.96 mph)	45 kmph (27.96 mph)
Gear Down Protection Light Load Vehicle Speed	11430	28	Min: 19 Max: 56	mph	45 kmph (27.96 mph)	45 kmph (27.96 mph)
Idle Speed Control						
Adjustable Low Idle Speed Enable (Read Only)	50512	Y		49	Y	Y
Low Idle Speed	10030	750	Min: 600 Max: 800	rpm	650	650
Idle Shutdown						
Idle Shutdown Enable	20051	N		Y = Enable N = Disable	Y	Y
Idle Shutdown Timer	10170	60.0	Min: 0.5 Max: 1440.0	Minutes	5	5
Idle Shutdown in PTO Enable	20071	N		Y = Enable N = Disable	N	N
Idle Shutdown Manual Override Enable	20081	N		Y = Enable N = Disable	Y	Y
Idle Shutdown service brake switch interaction	50057	N		Y = Enable N = Disable	Y	Y
Load-Based Speed Control						
Load Based Speed Control Enable	50016	N		Y = Enable N = Disable	Y	Y
Load Based Speed Control Low Engine Speed Breakpoint	50027	1800	Min: 1200 Max: 2100	rpm	1800	1800
Load Based Speed Control High Engine Speed Breakpoint	50028	1800	Min: 1500 Max: 2500	rpm	1800	1800
Load Based Speed Control Vehicle Mass Threshold Enable (Read Only)	50446	Y		Y = Enable N = Disable	Y	Y
Load Based Speed Control Vehicle Mass Threshold	50447	68343	Min: 22046 Max: 220460	lb	35000 kg (77161.79 lbs)	35000 kg (77161.79 lbs)
Road Speed Governor						
Road Speed Governor Enable (Read Only)	99001	Y		Y = Enable N = Disable	Y	Y
Maximum Accelerator Vehicle Speed	10000	56	Min: 12 Max: (13590)	mph	80 kmph (49.709 mph)	80 kmph (49.709 mph)
User Adjustable Fuel Economy						
User Adjustable Fuel Economy Enable	50437	N		Y = Enable N = Disable	Y	Y
UAFE Max Road Speed Enable	50475	Y		Y = Enable N = Disable	Y	Y
UAFE Max Road Speed Limit	50476	36	Min: 12 Max: 56	mph	79 kmph (49.09 mph)	79 kmph (49.09 mph)
UAFE Torque Limit Enable	50486	N		Y = Enable N = Disable	Y	Y
UAFE Mid Torque Limit Adjust Factor	50487	100	Min: 80 Max: 200	%	100	100
UAFE Max Torque Limit Adjust factor	50488	100	Min: 80 Max: 200	%	100	100
Accelerator Control						
Variable Rate Vehicle Acceleration Management	50040	N		Y = Enable N = Disable	Y	Y
Variable Acceleration 1	50041	0.6	Min: 0.0 Max: 8.1	mph/sec	1.5 kph/sec (0.93 mph/s)	1.5 kph/sec (0.93 mph/s)
Variable Acceleration 2	50042	0.6	Min: 0.0 Max: 8.1	mph/sec	1.5 kph/sec (0.93 mph/s)	1.5 kph/sec (0.93 mph/s)
Variable Acceleration Speed 1	50043	6	Min: 0 Max: 124	mph	20 kmph (12.43 mph)	20 kmph (12.43 mph)
Variable Acceleration Speed 2	50044	37	Min: 0 Max: 124	mph	60 kmph (37.28 mph)	60 kmph (37.28 mph)
Engine Brakes						
Engine Brake Control Enable	23021	N		Y = Enable N = Disable	Y	Y
Engine Brake Type	12790	0		0 = Exhaust Brake 2 = Engine Brake	2	2
Engine Brake Minimum Vehicle Speed	11540	0	Min: 0 Max: 35	mph	0	0
Engine Brake Delay Time	11542	0.0	Min: 0.0 Max: 4095.9	Seconds	0	0
Engine Brake Service Brake Activation Enable	21251	N		Y = Enable N = Disable	N	N
Engine Brake Clutch Pedal Activation Enable	21253	N		Y = Enable N = Disable	Y	Y
Cruise Control and Engine Brake Interaction Enable	20601	N		Y = Enable N = Disable	N	N
Start Maximum Engine Braking in Cruise Control	12410	3.1	Min: 0.0 Max: 6.2	mph	--	--
Start Minimum Engine Braking in Cruise Control	12420	2.0	Min: 0.0 Max: 6.2	mph	--	--
Accelerator Base						
Clutch Switch	10790	0		0 = Not Installed 1 = Installed	1	1
Starter Lockout Enable	24000	N		Y = Enable N = Disable	Y	Y
Starter Lockout Relay Type	22161	0		0 = Normally Closed 1 = Normally Open	1	1
Transmission Setup						
Transmission Top Gear Ratio	10300	1.000	Min: 0.000 Max: 64.000	Ratio	1	1
Gear Down Transmission Ratio	11340	1.330	Min: 0.000 Max: 64.000	Ratio	1.33	1.33
Transmission Type	11410	0		0 = Manual ; 1 = Automatic ; 2 = Fully Automated with Clutch ; 3 = Partially Automated 4 = Fully Automated without Clutch	0	0
Cruise Control Input						
Cruise Control	20021			Y = Enable N = Disable	Y	Y
Cruise control lower droop	10880			mph	0 kmph (0 mph)	0 kmph (0 mph)
Cruise control upper droop	10870			mph	0 kmph (0 mph)	0 kmph (0 mph)
Cruise control maximum vehicle speed	10090			mph	80 kmph (49.71 mph)	80 kmph (49.71 mph)
Cruise control auto resume	20611	N		Y = Enable N = Disable	N	N
Cruise control set speed save	21491	Y		Y = Enable N = Disable	Y	Y
Cruise Control and Engine Brake Interaction	20601	N		Y = Enable N = Disable	N	N
Switch Usage	10100	1		mph	1	1

Annexure-6						
Vehicle Speed Input						
Rear Axle Ratio	10050	5.120	Min: 15 Max: 15.990		5.12	5.12
Tire Size	10060	500	Min: 301 Max: 1287	Rev/Mile	295 rev/km (475 rev/mile)	295 rev/km (475 rev/mile)
Number of Transmission Tailshaft Gear Teeth	10070	4	Min: 1 Max: 64		4	4
Vehicle Speed Sensor Type	10080	0		(0 = None) ; 1 = Magnetic; 3 = Data Link - Tailshaft	1	1
Maintenance Monitor						
Maintenance Monitor Enable	20131	N		Y = Enable N = Disable	Y	Y
Maintenance Monitor Alert Percentage	10860	90	Min: 0 Max: 100	%	90	90
Maintenance Monitor Operating Mode	10230	0	0=Automatic ; 1=Time 2=Fuel ; 3=Distance	0 = Automatic ; 1 = Time 2 = Fuel ; 3 = Distance	3	3
Maintenance Monitor Time	10290				1000 sec	1000 sec
Maintenance Monitor Distance	10280	25000	Min: 0 Max: 1000000	Miles	40000 km (24854.848 miles)	40000 km (24854.848 miles)
Maintenance Monitor Fuel	10250				2404	2404
Alternator Failure Warning	22571	N		Y = Enable N = Disable	Y	Y
Anti-Tampering Measures						
Maximum Engine Speed Without Vehicle Speed Source	10020	2850	Min: 900 Max: 3000	rpm	1850	1850
Vehicle Speed Sensor Anti-Tampering Enable	20761	N		Y = Enable N = Disable	Y	Y
Engine Protection						
Engine Protection Shutdown Enable	20091	Y		Y = Enable N = Disable	Y	Y
Service Brake						
Service Brake Switch Enable	50070	N		Y = Enable N = Disable	Y	Y
Fan Control						
Air Conditioning Pressure Switch Input	21712	Disable		Y = Enable N = Disable	Y	N
Min Fan ON Time for Air Conditioning Pressure Switch	11350			Seconds	5	---
Fan Type	12490				7	7
Fan drive ratio	50670			ratio	1.2	1.2
Air Condition Control (Applicable only for SIGNA cabin)						
Air Conditioner Control	51260				Y	N
Air Conditioner On Idle Speed Adjust	51262				N	---
Powertrain Protection (PTP)						
Powertrain Protection	21781	Disable		Y = Enable N = Disable	N	N

DEBABRATA DAS [TMCV, Engineering, Jamshedpur]

From: AJAYKUMAR JADHAV [TMCV, Engineering, Pune]
Sent: 08 March 2023 12:58
To: SWAPNIL JADHAV [TMCV, Engineering, Pune]; GURMEET SINGH [TMCV, Engineering, Pune]; ANIRUDDHA TIRUKHE [TMCV, Engineering, Pune]; SANDIP PHAPALE [TMCV, Engineering, Pune]; DEBABRATA DAS [TMCV, Engineering, Jamshedpur]
Subject: M&HCV Summer Trials

Dear Team,

PFA trailing mail for summer trials are not required for 5.6 and 6.7L engine.

Regards,

Ajay

From: AJAYKUMAR JADHAV [TMCV, Engineering, Pune]

Sent: 08 June 2022 17:18

To: 'Raj Mohanty' <raj.mohanty@cummins.com>; UDAY HADAP [ERC, Engineering, Pune] <uday.hadap@tatamotors.com>; SUBHAYU DAS [TMCV, Engineering, Jamshedpur] <subhayu.das@tatamotors.com>; SANJEEV KUMAR [TMCV, Engineering, Jamshedpur] <sanjeev1.kumar@tatamotors.com>

Cc: Abhijit M Chittewan <abhijit.chittewan@cummins.com>; G. Karthikeyan <karthikeyan.ganesan@cummins.com>; Madhurima Biswas <madhurima.biswas@cummins.com>; PRAHLADA ANAND [ERC, Engineering, Pune] <anand.p@tatamotors.com>; SANJEEV KUMAR [TMCV, Engineering, Jamshedpur] <sanjeev1.kumar@tatamotors.com>; Abhijit M Chittewan <abhijit.chittewan@cummins.com>; G. Karthikeyan <karthikeyan.ganesan@cummins.com>; Sachin Solanki <sachin.solanki@cummins.com>; Gupta Deepak <gupta.deepak@cummins.com>; VISHNU DIXIT [TMCV, Engineering, Jamshedpur] <vishnud@tatamotors.com>; SANDIP GAIDHANE [TMCV, Engineering, Pune] <sandip.gaidhane@tatamotors.com>; MAHESH PISE [TMCV, Engineering, Pune] <mahesh.pise@tatamotors.com>; VINODKUMAR S M [TMCV, Engineering, Pune] <vinodkumar.sm@tatamotors.com>; Rahul Singh (TTL) <rahuls4.ttl@tatamotors.com>

Subject: RE: M&HCV Summer Trials

Dear All,

As per trailing mail confirmation from Cummins team. Only LPC trials is planned on 4830.TK as worst case at JSR and summer trials are not required for 5.6 and 6.7L engine.

Subhayu/ Sanjeev Sir: Kindly confirm 48T vehicle availability for LPC trials at JSR . Cummins team will join trials along with instrument setup.

Thanks for confirmation.

Regards,

Ajay

From: Raj Mohanty <raj.mohanty@cummins.com>

Sent: 02 June 2022 15:58

To: UDAY HADAP [ERC, Engineering, Pune] <uday.hadap@tatamotors.com>

Cc: Abhijit M Chittewan <abhijit.chittewan@cummins.com>; G. Karthikeyan <karthikeyan.ganesan@cummins.com>; Madhurima Biswas <madhurima.biswas@cummins.com>; PRAHLADA ANAND [ERC, Engineering, Pune] <anand.p@tatamotors.com>; SANJEEV KUMAR [TMCV, Engineering, Jamshedpur] <sanjeev1.kumar@tatamotors.com>; AJAYKUMAR JADHAV [TMCV, Engineering, Pune] <ajaykumar_jadhav@tatamotors.com>; Abhijit M Chittewan <abhijit.chittewan@cummins.com>; G. Karthikeyan <karthikeyan.ganesan@cummins.com>; Sachin Solanki <sachin.solanki@cummins.com>; Gupta Deepak <gupta.deepak@cummins.com>

Subject: RE: M&HCV Summer Trials

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Dear Uday,

LPC trial is planned in Signa 4830.TK vehicle at JSR as a worst case based on higher fuel layout.

Regards,

Raj Mohanty

Application Engineering

+91 90310 56223

From: uday.hadap@tatamotors.com <uday.hadap@tatamotors.com>

Sent: 30 May 2022 15:07

To: Gupta Deepak <gupta.deepak@cummins.com>

Cc: anand.p@tatamotors.com; sanjeev1.kumar@tatamotors.com; ajaykumar_jadhav@tatamotors.com; Abhijit M Chittewan <abhijit.chittewan@cummins.com>; G. Karthikeyan <karthikeyan.ganesan@cummins.com>; Sachin Solanki <sachin.solanki@cummins.com>

Subject: M&HCV Summer Trials

Importance: High

EXTERNAL SENDER: This email originated outside of Cummins. Do not click links or open attachments unless you verify the sender and know the content is safe.

Dear Deepak,

Reference discussion last week. Kindly confirm Cummins team availability for summer trials (Jaisalmer) with instrument & include the tests to be conducted.

Engine	Model	Responsible person	Location	Remark	Cummins team remark if any
5.6L	Signa 3023.T	Uday Hadap	Jaisalmer	Discussion under progress (Details will be shared shortly by Engine) Note: LPC trials not required for summer trials . final confirmation required from engine if any additional tests with supplier	
6.7L	Signa 5530.S	Uday Hadap	Jaisalmer		

Regards,

Uday

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